Perceived versus Calibrated Income Risks in Heterogeneous-Agent Consumption Models

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Abstract

Heterogeneous-agent consumption–saving models typically calibrate idiosyncratic income risk from unexplained income dispersion, yet these measures differ from household perceptions. Using data from the New York Fed's *Survey of Consumer Expectations*, I directly calibrate perceived income risks and show they are lower and more heterogeneous than conventional estimates. This "perception wedge" arises from unobserved heterogeneity and under-perception of risk (overconfidence). Incorporating subjective beliefs into a heterogeneous-agent life-cycle model substantially improves the fit to observed wealth inequality and the prevalence of hand-to-mouth households. The results highlight the importance of the "choice channel," whereby subjective risk perceptions shape wealth accumulation dynamics.

Keywords: Income risks, Incomplete market, Perceptions, Precautionary saving, Heterogeneous-agent model, Life cycle

JEL Codes: D14, E21, E71, G51

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1. Introduction

Income risks matter for both individual behavior and macroeconomic outcomes. Given identical expected income and homogeneous risk preferences, different degrees of risk lead to different savings, consumption, and portfolio choices. This is well understood in models in which either the prudence in the utility function (Kimball 1990; Carroll and Kimball 2001), or occasionally binding constraints induce precautionary savings. It is widely accepted, based on empirical research, that idiosyncratic income risks are at most partially insured (Blundell et al. 2008) and that such market incompleteness leads to ex-post wealth inequality (Aiyagari 1994; Huggett 1996; Carroll and Samwick 1997; Krusell and Smith 1998) and different degrees of the marginal propensity to consume (MPC) (Krueger et al. 2016; Carroll et al. 2017). This also changes the mechanisms by which macroeconomic policies can affect economic outcomes (Krueger et al. 2016; Kaplan et al. 2018; Auclert 2019). Furthermore, aggregate movements in the degree of idiosyncratic income risks can drive time-varying precautionary savings motives—another source of business cycle fluctuations (Challe and Ragot 2016; McKay 2017; Heathcote and Perri 2018; Kaplan and Violante 2018; Den Haan et al. 2018; Bayer et al. 2019; Acharya and Dogra 2020; Ravn and Sterk 2021; Harmenberg and Öberg 2021).

While idiosyncratic income risks are one of the key inputs in the class of macroeconomic models featuring market incompleteness, their size and heterogeneity are not directly observable. One common practice in this literature is inferring risks from unexplained dispersion in income changes seen in panel data, and then treating the estimates as the true degree of risks as perceived by the agents who make decisions in the model (see, e.g., Krueger et al. 2016; Bayer et al. 2019; Kaplan et al. 2018). However, this estimation practice has limitations.

The method economists use to calibrate the size and persistence of income risks, as perceived by the agents, is subject to problems such as those caused by unobserved heterogeneity or model misspecification. Certain information—either an individual's ex-ante heterogeneity or advance information about future income or risks—enters an agent's information set from time to time but is not directly observable by economists. Meanwhile, agents may subjectively perceive risks differently from their objective counterparts for various reasons, such as incomplete information and overconfidence. If the risks economists calibrate based on flawed estimations differ from those the agents perceive, then the model's implications will fail to match the agents' behavior even if the model itself is otherwise right.

This paper addresses this issue by utilizing the perceived risks of labor income surveyed by the New York Fed's *Survey of Consumer Expectations* (SCE). Building on the previous work that focuses on estimating partial insurance using expectational surveys (Pistaferri 2001; Kaufmann and Pistaferri 2009), this paper uses SCE's density survey and expected job transition probabilities, which directly reveal perceived income risks. In the density survey, respondents are asked to provide histogram-type forecasts of their wage growth over the next 12 months; they also report their perceived job-finding and separation probabilities. When the individual density forecast is available, a parametric density distribution can be fit to obtain the individual-specific subjective distribution. Then, the second moment, the implied variance of the subjective distribution, allows me to directly characterize the perceived risk profile without relying on external estimates from cross-sectional microdata. This provides a direct measure of the risk perception

that presumably guides individual decisions.

With the individual-specific reported perceived risks (PR) in hand, I first confirm that the differences in the mean risks across groups (age, gender, education, etc.) measured by the conventional method do capture some between-group differences in the mean self-reported perceptions (e.g., low-income young females are measured as, and perceive themselves as, facing higher risks than middle-aged middle-income males). However, the patterns of the two do not always align; for instance, perceived risks, unlike calibrated risks, decrease with education level. These patterns align well with those reported by Koşar and Van der Klaauw (2025), who measure perceived risks in the SCE by the interquartile range. More importantly, within every such group, there is also a great deal of heterogeneity in the PR that is not captured using the conventional approach. The R^2 from regressing the PR on the conventional explanatory variables does not exceed 0.1, indicating that the traditional method fails to capture at least 90 percent of the heterogeneity in the perceived risks. As with wage risks, I find that perceptions of unemployment risk are widely heterogeneous.

The reported perceptions matter only if they are meaningfully linked to household behaviors—and I confirm that they are linked to households' consumption plans. In particular, using SCE data, I find that an individual's expected spending growth is positively associated with their own perceived income risk, alongside other forces implied by a non-linear Euler equation. This association provides ex-ante evidence of precautionary saving motives at the micro level. My analysis underscores that, because the theoretical prediction is originally about an ex-ante correlation between perceived risk and planned consumption, precautionary saving motives are actually better tested with ex-ante expected consumption growth instead of ex-post outcomes, and based on the same individuals' expectations instead of their indirect proxies.

Another major finding of the paper that turns out to carry rich implications is that the perceived wage risks, on average, are *lower* than the indirectly calibrated size of the risks, even within groups. Specifically, the perceived annual real wage risk is around 3%-4% in terms of standard deviations, while the estimate based on panel data of wage growth following the conventional approach is at least 10%, combining the size of the permanent and transitory shocks. I confirm that this finding is robust to alternative specifications of the wage process, different frequencies of shocks, and the most conservative lower bound of the external estimates on wage risks, such as those by Low et al. (2010), in addition to other broader-based income measures in the existing literature (Table A.4). This finding is corroborated by a closely related contemporaneous study by Caplin et al. (2023), who also show that survey-reported earnings risks are lower than their indirectly estimated counterparts that use Danish administrative records.

It is also found that the wedge between perceived and calibrated risk arises from two sources that are both important. The first is unobserved heterogeneity, in that agents hold private information regarding their wage growth rate that is misattributed to shocks by econometricians. The second is an under-perception of risk relative to the objective income process taken as true, which I refer to as overconfidence. In particular, up to about 4 percentage points within the wedge reflects the expected heterogeneity in yearly wage growth rates by the agents, which is potentially private knowledge unavailable to econometricians, and the rest, about 20 percentage points, reflects overconfidence. Combining survey expectations, perceived risks, and realized wage growth allows me to decompose the wedge into these two components. The decomposition is conditional on

the objective income process taken as true and on interpreting expectational dispersion as genuine heterogeneity rather than biased expectations. Intuitively, the expected dispersion of agents reflects heterogeneity, the unexpected dispersion reflects the objective risk, and the residual gap between the objective risk and the perceived risk measures the under-perception of risk that I call overconfidence. I also consider the case where expectational dispersion of wage growth rates contains idiosyncratic biases, implying a wider expectational dispersion than the true heterogeneity. Regardless, it remains that both unobserved heterogeneity and overconfidence play a role in driving the wedge.

This evidence motivates me to utilize survey-implied risks, as measures of what agents truly perceive, to calibrate income risks in a standard partial-equilibrium, incomplete-market, life-cycle, one-asset consumption-saving model to quantify these effects on liquid asset accumulation and the MPC. The baseline model blends the work of Huggett (1996), the income structure of Carroll and Samwick (1997), and persistent unemployment spells and unemployment benefits à la Krueger et al. (2016) and Carroll et al. (2017). In the preferred recalibrated subjective model, I instead use survey PRs to calibrate subjective perceived wage and unemployment risks, and also add the ex-ante different life cycle growth profiles to be consistent with the identified heterogeneity. The model is subjective in the sense that perceptions and actual income processes are separately calibrated. Compared with the baseline, the recalibrated subjective model markedly improves the match to the distribution of liquid wealth. It raises the share of households with less than half a month of income in liquid assets, the hand-to-mouth (H2M) population, from 0.09 to 0.31 (data: 0.35). It also increases the wealth Gini from 0.73 to 0.84, bringing it closer to the empirical value of 0.89. In addition, the model-implied MPCs increase significantly by 8 percentage points to 0.25, consistent with the stated MPCs in the same survey. What is particularly notable is the life-cycle profile of H2M agents, which closely mirrors the data.

Three forces together drive the model closer to the data. First, heterogeneity in perceived wage and unemployment risks increases asset inequality by inducing heterogeneity in consumption/saving choices. Second, a smaller perceived risk than in the baseline model, i.e., overconfidence, implies reduced precautionary saving motives, hence a lower level of wealth accumulation by all agents in the economy. Among the changes to perceived wage risks, it is chiefly this channel that raises the model's hand-to-mouth share. Third, a lower degree of perceived risks implies a higher degree of predictable heterogeneity in wage growth, which translates to differential asset accumulations. A degree of ex-ante heterogeneity in the life-cycle permanent earning growth profiles with a dispersion of 2.8 percentage points per year, a seemingly small number, actually implies significantly increased permanent income heterogeneity in the model.[1]

Since the survey expectations on wage growth and job transitions are separately elicited, I can also quantify the relative importance of perceived wage risks and unemployment risks in the improvement of the model fit. The recalibration of wage risks accounts for the bulk of the higher wealth inequality: introduced alone, it delivers 8 points of the 11-percentage-point increase in the wealth Gini, versus 2 for unemployment risks alone. Meanwhile, the two channels are strongly complementary in accounting

[1]This echoes several studies that emphasize the role of ex-ante heterogeneity in income processes, especially in growth rates of earnings, in addition to ex-post realization of income shocks, in accounting for income inequality: Guvenen (2007), Primiceri and Van Rens (2009), and Karahan et al. (2022).

for the H2M share: individually they raise it by 9 and 6 percentage points, respectively, but jointly by 22. Intuitively, a household maintains a liquid buffer as long as either perceived risk remains substantial, so it is only when both wage and unemployment risks are perceived to be low that precautionary saving collapses and hand-to-mouth status becomes optimal.

My benchmark subjective model deviates from the FIRE (full-information rational-expectation) assumption in that the perceived risk determining consumption/saving decisions in the model differs from the actual income risk parameters governing draws of income shocks.[2] In alternative models, I set the actual risk parameters equal to the perceived risks, treating perceptions as the model's truth. The difference between these variants and the subjective model also reveals a breakdown of the model implications into two channels: one via ex-ante savings behavior resulting from risk perceptions, or the "choice" channel, and the other via ex-post realized income inequality, or the "outcome" channel.

The main finding from this extension is that the "choice" channel is the key: whether the actual risks align with perceptions or with the conventional calibration, letting consumption/savings decisions be driven by the survey-reported risks alone is what yields a closer match of the model with the empirically measured wealth inequality and the fraction of low-liquid-asset-holding consumers. This reinforces a message that is echoed by many other studies that are based on expectation surveys: directly reported perceptions, albeit possibly subjective, still better explain the behaviors of heterogeneous agents and generate more realistic downstream macroeconomic implications than indirectly calibrated expectations that often rely on strong assumptions.

Finally, the paper demonstrates that the superior fit of the subjective model is robust to, and often complements, a range of additional features discussed in the literature. First, I consider the existence of expectational biases, which lead to dispersion in subjective life-cycle income profiles that is wider than that in their actual counterparts. Second, I allow for different subjective life-cycle unfolding patterns of growth-rate heterogeneity, assuming that agents perceive the dispersion of growth rates as monotonically increasing, decreasing, or hump-shaped over the life cycle. The main predictions of the subjective model are largely preserved under these alternative specifications. I further show that introducing preference heterogeneity—particularly non-homothetic preferences (Straub 2019)—and relaxing the zero-borrowing constraint improves the model's empirical fit. Moreover, I document that the subjective model delivers better performance across various moments when comparing preference calibrations that are indirectly inferred by targeting different sets of moments.

Related literature

This paper relates to several strands of the literature. First, following Dominitz and Manski (1997) and Manski (2004, 2018), it uses survey data to study not only subjective income expectations but also perceived income risks. I compare these perceptions with income risks estimated from panel data using standard methods (MaCurdy 1982; Abowd and Card 1989; Gottschalk et al. 1994). While most of the income-risk literature focuses on the time-series, cyclical, and cross-sectional properties of estimated risks

[2]There is mounting evidence in macroeconomics that people form expectations in ways that deviate from the FIRE. See, for example, Mankiw et al. (2003), Reis (2006), Coibion and Gorodnichenko (2012), and Wang (2024). However, most of this type of evidence is based on macroeconomic expectations, such as those of inflation.

(see, among others, Meghir and Pistaferri 2004; Storesletten et al. 2004; Blundell et al. 2008; Guvenen et al. 2014; Arellano et al. 2017; Guvenen et al. 2019; Bayer et al. 2019; Guvenen et al. 2021; Bloom et al. 2018; Moffitt 2020), this paper emphasizes the discrepancy between perceived and indirectly estimated risks. A closely related literature studies job-separation and job-finding expectations (Stephens Jr 2004; Drahs et al. 2018; Conlon et al. 2018; Mueller et al. 2021; Mueller and Spinnewijn 2021; Du et al. 2024). In particular, related to Balleer et al. (2021), which documents over-optimistic and heterogeneous job-finding beliefs in a heterogeneous-agent model, this paper extends the focus to perceived wage risks and argues that lower perceived risks may reflect not only behavioral biases but also the presence of advance information. For unemployment risk, this paper emphasizes the substantial heterogeneity in perceived risks across households.

The paper most closely related to this one is the contemporaneous study by Caplin et al. (2023). A key methodological difference concerns how subjective risks are compared with their conventional counterparts. Caplin et al. (2023) simulate unconditional earnings distributions from surveyed beliefs and compare them with Danish cross-sectional administrative data, whereas this paper estimates conditional income risks using standard methods and compares them with conditional risk perceptions reported in the survey. Despite these differences, both studies find that perceived earnings risks are lower than those inferred indirectly, consistent with unobserved heterogeneity and overconfidence. The two papers also differ in their quantitative frameworks: this study uses a standard life-cycle incomplete-markets model, building on Carroll et al. (2017) to study liquid wealth accumulation, while Caplin et al. (2023) adopt a search-and-matching model.

Second, this paper relates to the "insurance versus information" issue in the partial insurance literature (Pistaferri 2001; Kaufmann and Pistaferri 2009; Meghir and Pistaferri 2011; Kaplan and Violante 2010; Hendren 2017; Stoltenberg and Uhlendorff 2022). Empirical estimates of consumption insurance often conflate unanticipated income shocks with information already known to agents, leading to excess smoothness (Flavin 1988). Like this literature, I use surveyed expectations to measure agents' conditional beliefs. The key differences are that I directly use survey-reported income risks from density forecasts rather than inferring risks from forecast errors, and, crucially, that I emphasize that the survey perceptions may reflect not only superior information but also overconfidence.

Third, this paper contributes to the literature on consumption, saving, and the macroeconomic implications of subjective income expectations. Prior work studies learning about income risk and heterogeneity in income profiles (e.g., Pischke 1995; Wang 2004; Guvenen 2007), information frictions in household expectations (Carroll et al. 2018), biased income beliefs (Rozsypal and Schlafmann 2023), and information choice in heterogeneous-agent models (Broer et al. 2021). This paper differs in two key ways: it focuses on the second moment of income, namely income risk, and it disciplines belief heterogeneity directly using survey data, remaining agnostic about the specific expectations-formation mechanism.

2. Theoretical framework

2.1. Wage process and perceived risk

To be consistent with the survey-elicited questions in the SCE, I separately focus on the wage risk conditional on employment status in this section and the risks associated with job transitions in a later section. Conditional on being employed in the same job, in the same position, and having the same work hours, the log idiosyncratic earnings, or the wage rate, of an individual i at time t, $w_{i,t}$, consists of a predictable component, $z_{i,t}$, and a stochastic component, $e_{i,t}$. (Equation 1)

$$w_{i,t} = z_{i,t} + e_{i,t} \tag{1}$$

There is an extensive discussion in the literature about the exact time-series nature of the stochastic component e. For instance, it may consist of both a permanent and a transitory component (MaCurdy 1982; Abowd and Card 1989; Gottschalk et al. 1994; Carroll and Samwick 1997; Blundell et al. 2008; Kaplan and Violante 2010). Or some of the literature may replace the permanent component with a stationary/persistent component in the form of an autoregressive (AR) process, as in Storesletten et al. (2004); Guvenen (2007, 2009). The transitory component could be moderately serially correlated following a moving-average (MA) process (Meghir and Pistaferri 2004). I first proceed with the generic structure, as in Equation 1, without differentiating these various specifications. I defer that discussion to Section 4.2.

The wage growth from t to $t+1$ consists of the predictable change in $z_{i,t+1}$ and the change in the stochastic component $e_{i,t}$.

$$\Delta w_{i,t+1} = \Delta z_{i,t+1} + \Delta e_{i,t+1} \tag{2}$$

Under the assumption of full-information rational expectation (FIRE), all of the shocks that are realized until t are observed by the agent at time t. Therefore, the expected volatility under the FIRE (with the superscript $*$) is the conditional variance of the wage growth from t to $t+1$.

$$Var^*_{i,t}(\Delta w_{i,t+1}) = Var^*_{i,t}(\Delta e_{i,t+1}) \tag{3}$$

The predictable changes do not enter the PR. Hence, the PR under FIRE is the *conditional* variance of the change in the stochastic component, $Var^*_{i,t}(\Delta e_{i,t+1})$. Notice that this crucially depends on the time-series nature of $e_{i,t}$.

Consider $Var^*_{i,t}(\Delta w_{i,t+1})$ as the FIRE counterpart of what this paper hereafter refers to as the subjective perceived risk (PR), which is denoted as $Var_{i,t}(\Delta w_{i,t+1})$ (without the superscript $*$) and is directly measured in the survey.

Economists do not directly observe the size of the true PR. To estimate it, researchers usually start by obtaining an approximation of the stochastic component, $e_{i,t}$, denoted as $\hat{e}_{i,t}$, by subtracting the approximated predictable change, $\Delta\hat{z}_{i,t}$, from the observed wage growth in the panel data, $\Delta w_{i,t}$, that is $\Delta\hat{e}_{i,t} = \Delta w_{i,t} - \Delta\hat{z}_{i,t}$. To mimic $z_{i,t}$ from the agent's point of view, $\hat{z}_{i,t}$ commonly includes factors such as age polynomials, gender, education, and occupation. Hence, $\hat{e}_{i,t}$ are, essentially, the residuals of the first-step

wage regression controlling for a limited number of observable variables measured in the panel data. Then the cross-sectional variance of $\Delta\hat{e}_{i,t}$ is the input for estimating the income risk. It is usually referred to in the literature as the "volatility" (see, e.g., Gottschalk et al. 1994; Moffitt and Gottschalk 2002; Sabelhaus and Song 2010; Dynan et al. 2012; Bloom et al. 2018).

$$Var_c(\Delta\hat{e}_{i,c,t}) = Var_c(\Delta w_{i,c,t} - \Delta\hat{z}_{i,c,t}) \tag{4}$$

Note that common practice usually estimates income risks at the group level, denoted as c (such as age, education, and cohort), although, in theory, the risks as perceived by a FIRE agent could be totally individual-specific. This is because, at the individual level, there are no repeated realizations of the risks; only a particular realization of the shock is drawn (Equation 4). The within-group cross-sectional variation of a sufficiently large group size is needed for such an estimation.

Unlike the agent's PR, $Var_c(\Delta\hat{e}_{i,c,t})$ is an *unconditional* variance at the group level. It is crucial to make a distinction between the agent's *conditional* PR and the *unconditional* volatility that economists approximate. Two important issues affect the comparability of the two.

First, it is very likely that what is controlled for in the first step of the income regression, namely $\hat{z}_{i,c,t}$, does not perfectly coincide with what is *predictable* from the point of view of a FIRE agent. This is primarily because econometricians who have the earnings panel data cannot always control for the "unobserved heterogeneity", either purely ex-ante or due to wage innovations at time t, that is not measured in the data—the latter possibility is particularly relevant since the ex-ante heterogeneity can often be controlled for by individual fixed effects with panel data. This is often framed, from the point of view of the agents, as the "superior information" problem (Pistaferri 2001; Kaufmann and Pistaferri 2009). It refers to the possibility that agents have advance information regarding their wage growth, information that is not available to econometricians. For instance, a worker might be concerned that a recent dispute with their boss may negatively affect their wage the next year, but econometricians have no way of knowing this.

Second, the comparison is sensitive to the time-series nature of $e_{i,c,t}$. Again, this occurs because the economists' estimated volatility is unconditional, while the perception is conditional on the information until time t. To illustrate this point, imagine a very persistent component in the income shock. Under the aforementioned process, the estimated income volatility also includes the variance of the realized shock until t, which has already entered the agent's information set. Therefore, even if the econometricians perfectly recover the $e_{i,t}$ in the first-step regression, the presence of a persistent component in the income changes would result in differences between the PR and the estimated income volatility. Therefore, to approximate the true PR from the point of view of the agents, economists would need to recover a conditional variance using information from the unconditional variance, typically by assuming a particular time-series structure of the stochastic component e and using cross-sectional moment restrictions to estimate its size. I return to this discussion in Section 4.2.

To summarize, for two reasons, the survey-elicited PR has an invaluable use and is preferable to a conventional income risk estimation based on cross-sectional realizations, which is also used to parameterize macro models. First, survey-reported PR is, by

construction, conditional on each agent's information set, i, which is likely to include the intrinsic heterogeneity specific to the individual or the advance information useful for forecasting that individual's own wage growth. Economists who try to approximate the PR cannot do as well as the agents who answer the questions because the latter's information is not necessarily available to economists. Second, the survey-implied PR provides direct identification of the degree of heterogeneity of the income risk across individuals in the economy. This prevents modelers from possibly making imperfect assumptions when they estimate group-specific income risks by grouping individuals based on very limited dimensions of observable factors, such as education and age.

Importantly, for survey-reported PRs to outperform calibrated risks in predicting household behavior, it is sufficient that agents base their consumption-saving decisions on their perceptions—even if those perceptions deviate from true risks and violate FIRE, for instance, due to overconfidence. While the remainder of the paper evaluates the relevance of these assumptions, the key takeaway is that survey-based PRs, even when misaligned with objective risks, provide a better basis for predicting individual decisions than conventional calibrated risks, as long as agents act upon such perceptions. This is indeed the case, as reported in Section 4.6.

3. Data, variables, and density estimation

3.1. Data on perceived risks

The data used for this paper is obtained from the core module of the *Survey of Consumer Expectations* (SCE), conducted by the New York Fed, a monthly online survey for a rotating panel of around 1,300 household heads. My sample runs from June 2013, when the density forecasts were first fielded, through March 2020, or 82 months. Respondents remain on the panel for up to 12 consecutive months. In particular, I use the SCE's density forecasts of the earnings growth distribution conditional on employment and the reported probabilities of job finding and separation to measure perceived wage risks and unemployment risks, respectively. This section primarily discusses issues related to perceived wage risks, and in Section 4.5, I separately examine unemployment risk expectations.

In particular, the SCE question that elicits the density forecasts is framed as follows: "Suppose that 12 months from now, you are working in the same job at the same place for the same number of hours. In your view, what would you say is the percentage chance that 12 months from now your earnings on this job, before taxes and other deductions, will increase by x%?" Respondents are asked to assign probabilities to 10 unequally spaced bins to report their subjective distributions, with the probabilities summing to one. Then, I fit the bin-based density forecast in each survey response with a parametric distribution.[3] The variance of the estimated distribution naturally represents an individual-specific perceived risk.

To account for wage risk in real terms, I further add the individual-specific inflation uncertainty estimated using the same procedure and the same individual's density forecasts of inflation in the SCE. This procedure is predicated on the assumption that agents regard individual wage growth and aggregate inflation as independent random variables. This assumption is not perfect. To alleviate the concern of a correlation

[3]This follows the approach employed by Engelberg et al. (2009) and researchers at the New York Fed (Armantier et al. 2017). Appendix A.1 documents in detail the estimation methodology and its robustness.

between wage and inflation expectations, my sample ends in March 2020, as noted above, excluding the COVID-19 pandemic and the subsequent inflation surge in the United States. Arguably, with low and stable inflation in this sample period, the correlation between the two was much weaker.[4] In addition, I use both the adjusted PR in real terms and the nominal PR for the empirical results below.

For this paper, separately measuring perceived wage and unemployment risks has several unique advantages, but it also results in certain limitations. One of the advantages is that it avoids misconstruing changes in earnings due to voluntary labor supply decisions as risks. Empirical work estimating income risks is often based on data from total earnings or even household income, in which voluntary labor supply decisions and other variations confound the true degree of the uninsured idiosyncratic risks. The survey-based measure used here is not subject to this problem. Another advantage is that the separately elicited conditional belief measures allow this paper to evaluate the patterns of the two sources of income risks found to be both important (Low et al. 2010; Guvenen et al. 2021), and to quantify their respective importance in driving consumption/saving decisions. One limitation, however, is that because the SCE question conditions on the same job and hours, the measure excludes earnings risk arising from hours variation. The measures are also annual, abstracting from the high-frequency earnings instability documented by Ganong et al. (2025).

Meanwhile, as the SCE does not elicit the beliefs regarding all earning distributions conditional on different job transitions, it is difficult to make a comparison between the full subjective income distribution and its calibrated counterparts. On this front, Caplin et al. (2023) provide highly complementary evidence that is based on such an approach. They elicit subjective job-transition probabilities and unconditional earnings distributions for each scenario of job transitions. This enabled them to combine these data into a holistic income distribution. The authors report similar findings to those in this paper.

3.2. Wage data

I examine longitudinal data on individual labor earnings from the 2014–2017 and 2018–2022 annual data collections of the *Survey of Income and Program Participation* (SIPP), which cover calendar years 2013–2016 and 2017–2021, respectively. Each collection of the SIPP is designed to be a nationally representative sample of the U.S. population. The interviews, conducted once a year, collect data on individuals' monthly earnings, hours of work, and other labor market outcomes. On average, each individual is surveyed for 33 months over multiple waves of the survey. This annual interview structure causes the "seam" issue documented by Moore (2008), whereby reported changes in the answers (e.g., on wage growth) within the survey waves are systematically smaller than the cross-wave changes. For the baseline estimation, I exclude the cross-wave earnings growth, which produces a lower-bound estimate of the wage risk—and therefore a conservative benchmark for the comparison with perceived risks that follows. Appendix A.3 gives a more in-depth inspection of this issue.

For the purpose of this paper, using the SIPP to estimate the wage risk has obvious advantages over other commonly used datasets, the most notable of which is the *Panel*

[4]Several papers found very limited pass-throughs from inflation expectations to wage/income expectations of individual households, such as Jain et al. (2024) and Hajdini et al. (2025). Particularly relevant is Koşar and Van der Klaauw (2025), who found that in the SCE, density-forecast based inflation uncertainty is weakly correlated with earning uncertainty.

Study of Income Dynamics (PSID). The SIPP contains information that allows me to work with wage changes conditional on staying in the same job with the same employer, thanks to its detailed records of job transitions and unique employer identifier. In contrast, the PSID only provides biennial records of labor earnings for the years since 1997. For the overlapping periods between the SIPP and the SCE, it is possible to make a direct comparison between the realized wage risks at annual frequencies and the ex-ante perceptions of the wage risk. This is particularly crucial if the wage risk is time-varying and dependent on macroeconomic conditions.

To ensure the comparability between the perceptions and the realized outcomes, I obtain the hourly wage of workers employed by the same employer by dividing the total monthly earnings from the *primary job* by the average number of hours of work for the same job for only those who stay with the same employer for at least 2 years. To identify job stayers, I follow the same approach as Low et al. (2010) and I impose five criteria. I only include (1) the working-age population between age 20 and 60; (2) private-sector jobs, excluding workers employed in government or other public sectors; (3) those remaining in the same job as the previous year; (4) those whose monthly wage rates are no greater than 10 times, nor smaller than 0.1 times, the individual's own average wage, after further trimming the top and bottom percentile of the wage distribution within each month; and (5) those who do not have days away from work without pay during the reference month. This leaves me with a monthly panel of about 200 to 1,100 individual earners for the sample period, March 2013–December 2021, or 7,774 distinct workers and 68,175 person-month observations in total. Appendix A.3 discusses in greater detail the data selection procedure and reports the summary statistics.

4. Basic facts about perceived income risks

4.1. First pass: perceived risk versus wage volatility

As a starting point, I first empirically compare the conditional perceived risks in the SCE with the unconditional wage volatility from the SIPP. The former corresponds to $Var_{i,t}(\Delta w_{t+1})$ as defined in Section 2. The latter corresponds to the wage volatility $Var_c(\Delta \hat{e}_{i,t+1})$. Here, I treat each group c as a particular individual. Thus, the wage volatility is essentially the variance across multiple years of wage growth realizations of the same worker. Figure 1 plots the distribution of the PRs across individuals and the distribution of wage volatility at the individual level that can be explained by observable demographic variables such as age, gender, and education.

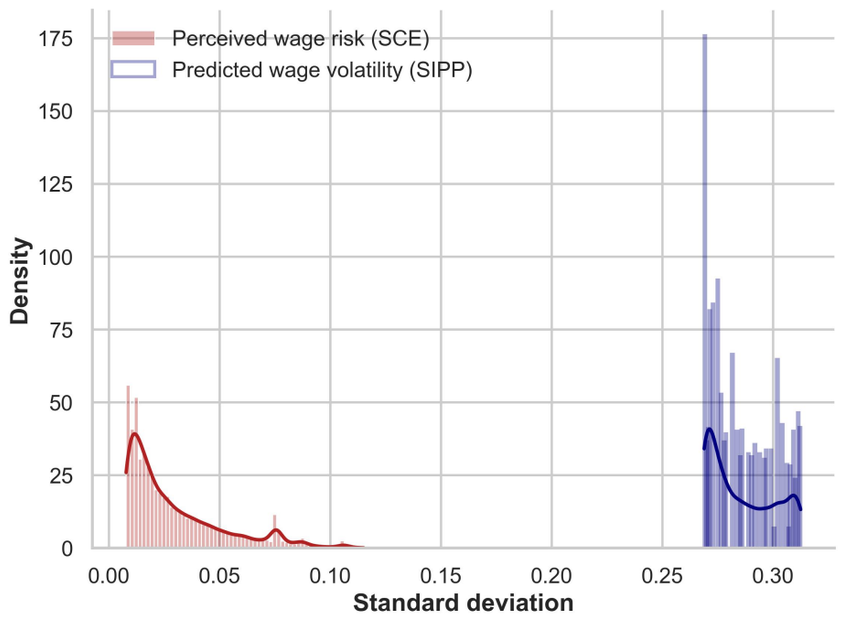
The first apparent pattern from the figure is that PRs are concentrated in a much lower range of values (2–4%) while, in contrast, the average predictable size of the wage volatility falls in the range of 25-30%. This indicates that across all workers, the realized year-over-year wage growth fluctuates with a much larger magnitude than perceived by workers, conditional on their respective information set. In Appendix Figure A.6, I validate this pattern with an alternative approach: comparing simulated wage growth rates based on density forecasts in the SCE with the wage growth distribution in the SIPP. The former has a significantly smaller dispersion than the latter, consistent with the finding here.

Given the discussion in Section 2, it is not surprising that perceived risks are lower than the wage volatility estimates since the latter is *unconditional*. However, I show in the following sections that the perceived risks remain well below the often-used

"calibrated risks", which are econometricians' proxies for the *conditional* risk from the point of view of the agents.

The second pattern in Figure 1 is that PRs display much greater heterogeneity than the dispersion of wage volatility explained by observables. As the figure shows, the distribution of PRs is substantially wider than the explainable component of individual wage volatility. This indicates that assuming wage volatility heterogeneity arises only from observed factors understates the full variation in perceived risks. Because wage volatility is the upstream input to risk calibration, this limited heterogeneity also carries over to the calibrated risks.

FIGURE 1. Dispersion in the perceived wage risk versus wage volatility



Note: Distributions of the PRs regarding the real wage growth in the SCE and the distribution of the individual wage volatility that is explained by the age, age polynomials, gender, education, and time fixed effects.

4.2. Calibrated risk based on decomposed shocks of different persistence

As previewed in Section 2, the calibrated risk estimated from realized panel data depends on assumptions made on the time-series properties of the wage shocks. Realized permanent (or persistent) shocks carry implications for future wages, whereas purely transitory shocks do not. Consequently, agents perceive different levels of risk depending on the nature of the shock. This distinction is essential for making a meaningful comparison between survey-reported perceived risks (PRs) and risks calibrated using standard econometric approaches.

To proceed, I adopt the permanent–transitory wage process referenced in Section 2, which is commonly used in a large body of literature. I specify that the stochastic component $e_{i,t}$ consists of a permanent component p that follows a random walk and a transitory component θ that is i.i.d. The shocks to both components are log-normally distributed, with mean zero and potentially time-varying variances σ^2_ψ and σ^2_θ (consistent with the specification in Equation 17).

$$
\begin{aligned}
e_{i,t} &= p_{i,t} + \theta_{i,t} \\
p_{i,t} &= p_{i,t-1} + \psi_{i,t}
\end{aligned}
\tag{5}
$$

Under this specific wage process, the PRs of a FIRE agent are equal to the sum of the variance of the two components $Var^{*}_{i,t}(\Delta e_{i,t+1}) = \sigma^2_{\psi,t} + \sigma^2_{\theta,t}$. But, in contrast, the wage volatility estimated from the panel data, provided that the change in the predictable component Δz is perfectly controlled for as in Equation 4, is a sample analog of $Var(\Delta e_{i,t}) = \sigma^2_{\psi,t} + \sigma^2_{\theta,t-1} + \sigma^2_{\theta,t}$. It differs from the PR by $\sigma^2_{\theta,t-1}$ precisely due to its unconditional nature. The intuition here is that the variance of the transitory shock that is realized at time $t-1$ is no longer perceived as the wage growth risk conditional on the information at time t.

Given this process, the calibrated risk, denoted as $\widehat{PR}$, is equal to the sum of the *estimates of* the permanent and transitory risks, $Var_t(\Delta \hat{e}_{i,t}) = \hat{\sigma}^2_{\psi} + \hat{\sigma}^2_{\theta}$. To obtain its magnitude, I follow the same generalized method of moments (GMM) estimation procedure as in the literature (Abowd and Card 1989; Carroll and Samwick 1997; Meghir and Pistaferri 2004; Blundell et al. 2008), whose approaches differ from one another only in minor ways depending on the model specification (see Appendix A.4.1 for details), to identify the time-averaged variances of the permanent and transitory components of the monthly wage growth, using the SIPP's wage data for the same period. I then convert these monthly risk parameters into annual frequencies, to be comparable to the perceived risks of the annual wage growth.[5]

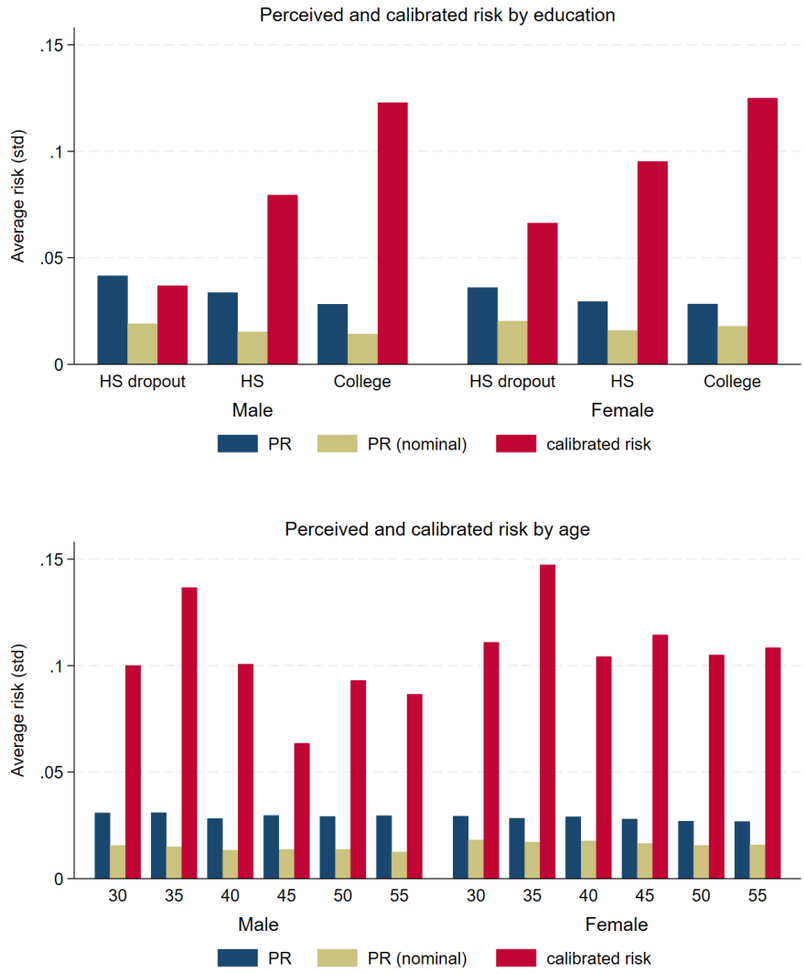
Table 1 reports group-specific estimates of total, permanent, and transitory wage risks from the wage panel, together with the corresponding average and median perceived risks. The estimated permanent wage risk, $\hat{\sigma}_{\psi}$, is about 10%, closely matching Low et al. (2010), who focus on wage rates of job stayers, excluding job-switching and unemployment spells. They likewise report annual permanent and transitory risks of roughly 10%. By contrast, my estimated transitory shocks are much smaller—below 3 percentage points—compared with the 10+ percentage point figures commonly cited based on changes in income/earnings at the annual frequency. This discrepancy mainly reflects my assumption of a higher frequency of shocks; consequently, volatility inferred from annual wage growth is not directly comparable to the sum of the calibrated permanent and transitory components. While the appropriate specification of high-frequency wage or income processes remains an active research question (Crawley et al. 2022; Crawley 2019; Druedahl et al. 2021), I adopt these estimates as calibration benchmarks. They provide a conservative lower bound against which perceived risks (PRs) can be meaningfully compared.

4.3. Perceived risks are heterogeneous

In both income risk estimation and the parameterization of incomplete market macro models, it is common practice to assume, first, that idiosyncratic risks differ as a function of certain observable factors such as education, gender, and age, and, second, that there

[5]The annual object is the point-in-time twelve-month wage change, matching what the survey elicits. The annual permanent variance is therefore the sum of the twelve monthly permanent variances over the forecast window, and the annual transitory variance is that of a single monthly transitory shock, since the current transitory draw is already known to the forecaster. Appendix A.4.2 provides the derivations for this result. Appendix A.4.3 provides alternative estimates for the quarterly and yearly frequencies.

FIGURE 2. Perceived risks versus calibrated wage risks by observable factors



Note: Real and nominal perceived risk (from the SCE), and calibrated wage risk (from the SIPP) of each education-gender (upper panel) or age-gender (bottom panel) group. The calibrated risk is equal to the estimated risk of the permanent and transitory components of the wage, based on the process specified in Equation 5.

is no additional within-group heterogeneity in the degree of the risk (see, e.g., Meghir and Pistaferri 2004; Sabelhaus and Song 2010; Bloom et al. 2018 on the empirical side, and the models of Cagetti 2003; Blundell et al. 2008; Carroll et al. 2017, which allow for heterogeneous risks across demographic variables). This section reports my finding that although the observed heterogeneity in the PR across individuals does reflect between-group differences along dimensions economists have commonly assumed, a dominant fraction of the cross-sectional differences in the PR cannot be explained by observables. Furthermore, even in those observable dimensions, the group heterogeneity seen in the PR does not coincide with that seen in the estimated risks.

Figure 2 plots the group average of the PRs (both in real and nominal terms), and the calibrated risk, $Var_{i,t}(\Delta\hat{e}_{i,t+1})$, based on an estimation of the specified wage process by age, gender, and education.

Regarding the education profile of wage risk, the calibrated risk is higher among more educated workers. This is consistent with the finding of Meghir and Pistaferri (2004), who examined total labor income rather than wage income. In contrast, risk perceptions exhibit the opposite pattern with respect to education level: less-educated workers report higher PRs than more-educated workers. Regarding the life-cycle pattern of risks, neither the wage volatility nor the estimated risks show a monotone pattern

over the life cycle. This age pattern of the wage risk is not necessarily contradictory to the well-documented declining pattern estimated using data on household income or total earnings (Cagetti 2003; Sabelhaus and Song 2010): it is likely that the decline in income risks over the life cycle has to do with non-wage risks or better insurance via work arrangements over the life cycle. In contrast, perceived risks almost monotonically decline over the life cycle for both males and females. These findings are confirmed in Table 1, which reports the group average PR, the wage volatility, and the estimated risks. These between-group differences are statistically significant, as shown by the PR regressions reported in Appendix Table A.1.

Even though PRs do differ across observed dimensions, such factors turn out to be unable to explain the majority of the cross-sectional dispersion in PRs. In particular, age, gender, and education jointly explain only 1.9% of the variance in perceived risks, and all of the observable factors in the SCE together explain at most 10%. Individual fixed effects alone deliver an R^2 of 67%, and adding every observable on top of them raises it by about one percentage point. Appendix A.2.1 plots the distribution of the unexplained residuals of the PRs, the expected wage growth, and the higher-order perceived risks such as the skewness after controlling for the observable individual characteristics, including the age, age polynomial, gender, education, type of work, and time fixed effects. All of these show sizable within-group heterogeneity.

This finding has two implications. First, the role of the within-group heterogeneity suggests that the conventional practice of estimating and modeling income risks as only differing by several observable dimensions fails to capture the dominant fraction of variations across individuals in wage risks. Second, the heterogeneity in the PR can be directly used to model the heterogeneous income risks without identifying the source of the heterogeneity. Therefore, in Section 5, my model calibration adopts such an approach.

4.4. Perceived risks are lower than calibrated risks

Another salient fact is that the PR is always *smaller* than the calibrated wage risk, as shown in Figure 2. In particular, the calibrated risk of different groups falls in the range of 4-14% per year in standard deviation units. In contrast, the average perceived risks reported in the survey are only about 3-4% and at least 50% smaller than the calibrated risks. For instance, a male high school graduate on average perceives his annual wage risk to be 4 percentage points, while the calibrated risk of the same group is above 9-10%.

Table 1 further confirms that within each group, the perceived risks (PRs) are systematically lower than the calibrated risks, even if the latter are economists' attempted proxy for the perceived risk. One natural explanation for this disconnect between perceived and calibrated risks seems to be unobservable heterogeneity or superior information. However, there is another possibility: agents may underperceive the true risks they face, therefore rendering PRs to be lower than the calibrated risk, even if the latter perfectly represents the conditional risk assessment of the agents under FIRE. Can the two explanations be distinguished? The next section delves into this question.

TABLE 1. Perceived risk, volatility, and calibrated risks by group

	PR(mean)	PR(median)	Volatility	CalibratedRisk	PermanentRisk	TransitoryRisk
Gender						
Male (50%)	0.031	0.024	0.356	0.103	0.097	0.0226
Female (49%)	0.03	0.024	0.397	0.113	0.106	0.027
Education						
HS dropout (0%)	0.036	0.021	0.359	0.052	0.05	0.0067
HS graduate (40%)	0.032	0.024	0.38	0.087	0.083	0.016
College/above (58%)	0.029	0.023	0.373	0.124	0.115	0.0311
5-year age range						
20 (2%)	0.038	0.032	0.382	0.069	0.068	0.0063
25 (12%)	0.033	0.028	0.359	0.135	0.132	0.0107
30 (13%)	0.031	0.025	0.338	0.104	0.096	0.0245
35 (14%)	0.031	0.024	0.338	0.141	0.128	0.0476
40 (13%)	0.03	0.023	0.433	0.102	0.093	0.0302
45 (14%)	0.029	0.022	0.37	0.09	0.085	0.0195
50 (14%)	0.029	0.021	0.351	0.099	0.095	0.0188
55 (15%)	0.029	0.02	0.434	0.098	0.092	0.023
Total (100%)	0.03	0.023	0.376	0.108	0.101	0.0248

Note: This table reports the square roots of the mean and median PRs ($Var_{i,t}(\Delta w_{i,t+1})$), the estimated annual wage volatility ($Var_c(\Delta\hat{e}_{i,t+1})$) for each group c, the calibrated risks ($\hat{\sigma}^2_\psi + \hat{\sigma}^2_\theta$), and the calibrated risks of the permanent ($\hat{\sigma}^2_\psi$) and transitory ($\hat{\sigma}^2_\theta$) wage components for different groups. Note that the calibrated risks are annualized based on estimates from monthly wage changes. See Table A.3 for calibrated risks with annual wage growth.

4.4.1. Unpacking the gap: unobserved heterogeneity or overconfidence

In this section, I present a framework to account for the gap between perceived risk and calibrated risk, and to empirically identify its components. Suppose the observed residual change in wage $\Delta\hat{e}_{i,t}$ by econometricians consists of two components. The first component $\Delta e_{i,t}$ is the truly unexpected wage shock to the agent. The second one, $\xi_{i,t}$, is a component anticipated by the agent at $t-1$ before its realization, but one that cannot be observed by econometricians. $\xi_{i,t}$ could stem from unconditional heterogeneity in wage growth rates across individuals that is private knowledge, or from innovations to i's wage growth between $t-1$ and t that enter the information set of i.

$$\Delta\hat{e}_{i,t} = \Delta e_{i,t} + \xi_{i,t} \tag{6}$$

To be consistent with the wage processes in the previous section, I also assume $\xi_{i,t}$ has the same time series property as $\Delta e_{i,t}$, consisting of a permanent component and a *change* in the transitory component.[6]

$$\xi_{i,t} = \xi^{\psi}_{i,t} + \Delta\xi^{\theta}_{i,t} \tag{7}$$

A good example of $\xi^{\psi}_{i,t}$, namely the individual-specific expected innovation to the permanent wage rate, is the wage rise expected by a fresh Ph.D. graduate who will start a professor's job the next year. An example of $\Delta\xi^{\theta}_{i,t}$, an expected transitory change that is unlikely to be observable by researchers, is the future earnings cut for a professor who expects to be on sabbatical leave for one semester.

To the extent that the respective components in $\Delta e_{i,t}$ and $\xi_{i,t}$ share the time-series

[6]This is similar to the specification of the unobserved heterogeneity in *income* as in Primiceri and Van Rens (2009), which only allows for a permanent component of the unobserved heterogeneity.

property, and they look identical to econometricians, it is easy to show that the common moment-based estimation as done in Section 4.2 (see Appendix A.4.1 for details) would yield a calibrated risk $Var_{i,t-1}(\Delta\hat{e}_{i,t}) = \hat{\sigma}^2_{\psi} + \hat{\sigma}^2_{\theta}$ that differs from the PR under FIRE by exactly the size of the variance of $\xi_{i,t}$.[7]

$$Var_{i,t-1}(\Delta\hat{e}_{i,t}) = Var^*_{i,t-1}(\Delta e_{i,t}) + \underbrace{\sigma^2_{\xi}}_{=\sigma^2_{\xi,\psi}+\sigma^2_{\xi,\theta}} \tag{8}$$

Except for the special case absent unobserved heterogeneity, i.e., $\sigma^2_{\xi} = 0$, the calibrated risk $Var_{i,t-1}(\Delta\hat{e}_{i,t})$ is certainly greater than the PR under FIRE $Var^*_{i,t-1}(\Delta e_{i,t})$.

Another possibility that rationalizes a lower PR than calibrated risk is that the perceived risk in the data $Var_{i,t-1}(\Delta e_{i,t})$ is smaller than $Var^*_{i,t-1}(\Delta e_{i,t})$ due to overconfidence, denoted by δ^2.

$$Var_{i,t-1}(\Delta e_{i,t}) = Var^*_{i,t-1}(\Delta e_{i,t}) - \delta^2 \tag{9}$$

Combining Equations 8 and 9 yields the following relationship between calibrated risk and survey-based PR—the two have a wedge that stems from either unobserved heterogeneity or overconfidence.

$$\underbrace{Var_{i,t-1}(\Delta\hat{e}_{i,t})}_{\text{Calibrated risk } \widehat{PR}} = \underbrace{Var_{i,t-1}(\Delta e_{i,t})}_{\text{Perceived risk } PR} + \delta^2 + \sigma^2_{\xi} \tag{10}$$

Without information in addition to the calibrated risk and perceived risk, one would be unable to separately identify the contribution of σ^2_{ξ} and δ^2. This is when the heterogeneity in reported expectations of wage growth $\mathbb{E}_{i,t-1}(\Delta w_{i,t})$, the cross-sectional dispersion of the first moment of the individual-specific wage growth distribution in the SCE, becomes useful. It provides an additional source of identification. Conditional on information as of $t-1$, the individual's expected wage growth rate $\mathbb{E}_{i,t-1}(\Delta w_{i,t})$ under FIRE is equal to $\xi_{i,t}$, as $\mathbb{E}^*_{i,t-1}(\Delta e_{i,t}) = 0$.

$$\mathbb{E}^*_{i,t-1}(\Delta w_{i,t}) = \xi_{i,t} \tag{11}$$

Under the assumption that reported expected wage growth $\mathbb{E}_{i,t-1}(\Delta w_{i,t}) = \mathbb{E}^*_{i,t-1}(\Delta w_{i,t})$, meaning that the individual i is aware of $\xi_{i,t}$ and there is no additional bias in expectations, the cross-sectional variance of expected wage growth rates across individuals identifies the variance of $\xi_{i,t}$. Intuitively, this means that if the respondents in the SCE do have private information about their wage growth rates, such private knowledge should also be reflected in their expectations about future wage

[7] To see this, rewrite $\Delta\hat{e}_{i,t} = \psi_{i,t} + \xi^{\psi}_{i,t} + \Delta\theta_{i,t} + \Delta\xi^{\theta}_{i,t}$, where the first two components combined are what econometricians would regard as a permanent shock to the wage and the latter two combined are the transitory shock. Then the respective risk estimates of the permanent and transitory shocks, namely $\hat{\sigma}^2_{\psi}$ and $\hat{\sigma}^2_{\theta}$, are upward biased relative to their true counterparts, σ^2_{ψ} and σ^2_{θ}, by exactly $\sigma^2_{\xi,\psi}$ and $\sigma^2_{\xi,\theta}$, which represent the variance of $\xi^{\psi}_{i,t}$ and $\xi^{\theta}_{i,t}$, respectively. Equation 8 follows upon defining $\sigma^2_{\xi} = \sigma^2_{\xi,\psi} + \sigma^2_{\xi,\theta}$.

growth.

$$\sigma_{\xi}^2 = Var(\mathbb{E}^*_{i,t-1}(\Delta w_{i,t})) \tag{12}$$

In the SCE, the cross-sectional standard deviation of reported expectations of wage growth is 0.04, as plotted in Figure 4D. It directly yields the estimate $\sigma_{\xi} = 0.04$. This implies, for each year, an average 4 percentage point dispersion across agents in their expected wage growth rate that might have to do with their private information, in the absence of expectational biases. With the σ_{ξ} identified, I can attribute the remaining wedge between calibrated and perceived risk in Equation 10 to the role of overconfidence δ^2.

Given the benchmark wage risk estimates of $\sigma_{\psi} = 0.15$ and $\sigma_{\theta} = 0.15$ (used to calibrate the baseline model in Section 5), hence a conventionally calibrated risk $\widehat{PR} = 0.21$ (in standard deviation units), the identification procedure yields the estimated unobserved heterogeneity $\sigma_{\xi} = 0.04$ and an overconfidence of $\delta = 0.20$. Correspondingly, the true risks σ_{ψ} and σ_{θ} are 0.147, slightly lower than the calibrated risks because the latter misattribute heterogeneity to risks.[8] Note here I make the most natural assumption that both heterogeneity σ_{ξ} and overconfidence δ comprise equal-sized contributions from a permanent and a transitory component. This assumption is needed because the SCE does not separately identify the perceived permanent and transitory components. Appendix Table A.5 shows that the model's improvement is essentially unchanged under alternative splits ($\kappa = 0.1$ and $\kappa = 10$). Regardless of this assumption, the estimates imply that a significant portion of the differences between calibrated risk and PR is due to agents underestimating the true risks—that is, overconfidence. I remain agnostic about the mechanism behind δ, which may be related to experience-based learning (Hertwig et al. 2004; Malmendier and Nagel 2011).

The estimated heterogeneity in wage expectations, σ_{ξ}, corresponds to a 4-percentage-point dispersion in expected wage growth and comprises both permanent and transitory components, $\sigma_{\xi,\psi}$ and $\sigma_{\xi,\theta}$. The extent of the permanent component crucially determines how heterogeneous life-cycle wage paths are across individuals. For example, even if only half of the total dispersion—about 2.8% per year—is due to permanent heterogeneity $\sigma_{\xi,\psi}$ that accumulates into lifetime permanent earnings, then the implications for earning profiles are large: an individual in the top third of the income distribution reaches a peak earning level around eight times their wage at age 25, whereas an individual in the bottom third, at their trough, earns only about half of their initial wage. In Section 5, I use this magnitude to calibrate heterogeneous wage growth paths, which are depicted in Figure A.8B. The heterogeneity in life-cycle earnings growth is in a similar range of values to those in Karahan et al. (2022) and Guvenen et al. (2021).

One potential concern with my estimates is that the expectations $\mathbb{E}_{i,t-1}(\Delta e_{i,t})$ may

[8] Using the wage risk estimates of Low et al. (2010), $\sigma_{\psi} = 0.10$ and $\sigma_{\theta} = 0.09$, or an implied calibrated risk $\widehat{PR} = 0.135$, yields a smaller estimate of overconfidence, $\delta = 0.13$. The corresponding true risks σ_{ψ} and σ_{θ} are 0.096 and 0.085, respectively. Applying the same identification to my own SIPP-based annual estimates from Table 1 ($\hat{\sigma}_{\psi} = 0.10$ and $\hat{\sigma}_{\theta} = 0.025$, an implied calibrated risk $\widehat{PR} = 0.104$) yields $\delta = 0.09$. The estimated overconfidence is therefore relative to the risk it is measured against: the lower the benchmark one adopts as the true risk, the smaller the residual left to attribute to under-perception. Even at the lowest of these three benchmarks, however, δ remains about three times the average perceived risk of 0.03.

contain idiosyncratic biases unrelated to true expectations. Formally, this can be written as

$$\mathbb{E}_{i,t-1}(\Delta e_{i,t}) = \mathbb{E}^{*}_{i,t-1}(\Delta e_{i,t}) + \varepsilon_{i,t} \tag{13}$$

where $\varepsilon_{i,t}$ captures individual bias. In this case, the estimate of σ^2_{ξ} is upward biased by exactly the variance of these biases. However, since the dispersion of expected wage growth in the data is at most 4%, the magnitude of such biases is likely negligible. Correcting for them would, in fact, raise the implied level of risk relative to the baseline estimates and would point to an even greater degree of overconfidence. I evaluate the quantitative implications of these alternative assumptions in the coming sections.

4.5. Unemployment risk perceptions

Since major employment status changes, such as job loss and finding, constitute another source of income risks in addition to wage fluctuations (Low et al. 2010; Davis and Von Wachter 2011), this section focuses on perceived unemployment risks (PUR). Unemployment risks are usually another central input of incomplete-market macroeconomic models (see Krueger et al. 2016; Bayer et al. 2019, among others). In these models, as in the approach to the wage risk, the common practice is to model the process of labor market transitions based on externally estimated stochastic processes, which are assumed to differ by at most one or two observed dimensions. This section shows that although the average survey expectations for job separation and finding probabilities are similar in magnitude to the measured labor-flow rates calculated from panel data (Fujita and Ramey 2009), this agreement on average masks substantial heterogeneity that is not commonly assumed in standard calibrations.

In particular, for different age, gender, and education groups, I compare the mean and median PUR in the SCE versus the actual group-specific employment transition rates. PURs are measured among SCE respondents by their reported expectations of the probability of losing their current job in the next 12 months and of finding a job in the next 3 months if the current job is lost. For realized transitions, I obtain the monthly and group-specific employment-to-unemployment and unemployment-to-employment flow rates, which are calculated from the *Current Population Survey* (CPS) and made available by the Federal Reserve Bank of San Francisco. For a fair comparison, I cast all rates into a 3-month horizon by (dis)aggregating monthly (12-month) flow rates to quarterly horizons.[9]

Figure 3 plots the comparisons. What is reassuring is that although the perceived and calibrated job risks are constructed independently of one another from different data sources, PURs' magnitudes, especially for job finding, are rather comparable to the realized job flow rates of the same group. For instance, the average perceived job finding probabilities range from 62% for the workers under the age of 25, to about 50% for workers older than 50, compared to a range of 58% to 50% from the young to the old. From high school graduates to college graduates, both perceived and actual job finding probabilities exhibit an increase from 50% to 60%. Regarding job separation, the between-group heterogeneity in perceived risks and realized job risks displays more

[9] Monthly probabilities $p_{t+j|t+j-1}$ compound to the quarter as $1-\prod_{j=1}^{3}(1-p_{t+j|t+j-1})$, assuming the event occurs at most once per quarter. An annual probability $p_{t+12|t}$ is disaggregated under a constant-hazard Poisson process: with $\lambda = -\ln(1-p_{t+12|t})$, the quarterly rate is $p_{t+3|t} = 1-\exp(-\lambda/4)$, so that compounding quarterly probabilities reproduces the annual rate.

distinct patterns. For instance, while actual job separation probabilities decline with education, the perception profile of education is rather flat. Due to the right-skewness, the mean perceived job separation probabilities are always higher than the medians. The actual job separation probabilities mostly lie between the two.

What is most notable from Figure 3 is the heterogeneity in PURs. As indicated by the wide 25/75 interquartile range around the median, individual respondents differ substantially in their estimates of separation and finding probabilities. For instance, the perceived job finding probabilities range from about 20% at their 25th percentile to 90% at their 75th. The perceived job separation rate ranges from zero at the 25th percentile to 5.5% at the 75th percentile. Because the survey question concerns individual-specific transitions, it is reasonable to assume that it reflects either unobserved heterogeneity or information about each individual's status, which economists cannot directly observe. Similar to wage risk heterogeneity, a regression of perceived job-finding and separation probabilities on all available individual-level controls yields R^2 values well below 10%. This suggests that the heterogeneity in calibrated risks is unlikely to capture all the heterogeneity revealed in the perceived risks.

It is worth noting that several studies do estimate substantial ex ante heterogeneity in the risks associated with job transitions. For instance, Karahan et al. (2022)'s estimates based on the SIPP panel suggest that the quarterly job separation probability ranges from 1% to 6%, and job finding probabilities range from 90% to 30% from top earners to bottom ones. However, such heterogeneity has rarely been calibrated in heterogeneous-agent models. Furthermore, these estimates are based on the inevitable assumption that risks differ along measurable dimensions (in their case, recent earnings), thereby falling short of characterizing the full scale of latent heterogeneity in risks. In contrast, perceived risks in survey data, which are directly measured at the individual level, can do so.

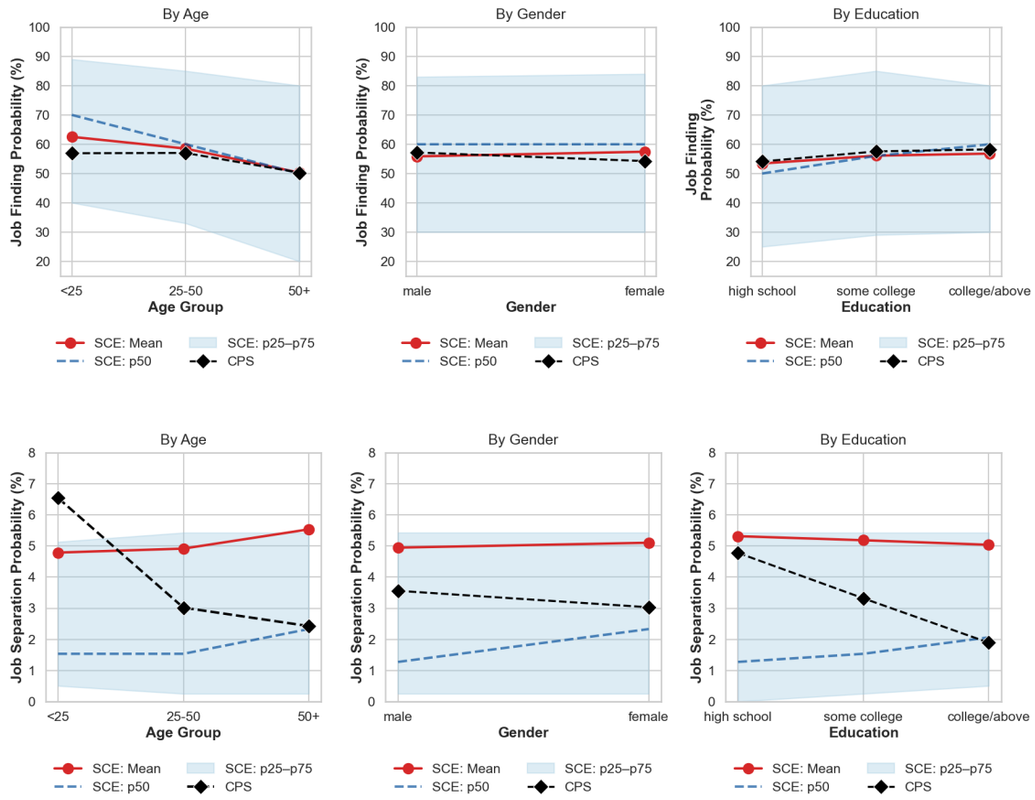
Recognizing that unemployment risks often fluctuate with macroeconomic conditions, Figure A.7 in the Appendix further compares the time-series patterns of perceived and calibrated unemployment risks. It again shows the substantial heterogeneity in perceived unemployment risks across households, although the average value tracks the realized job flow rates well at the aggregate level.

4.6. Perceived income risk and consumption plan

In this section, I show that perceptions of risks are systematically related to consumption plans. Due to precautionary savings motives, higher perceived risks should lead households to lower their current consumption, thus increasing their expected consumption growth. Despite such a clear directional prediction in theory, identifying the exact size of such an effect (i.e., the effect of perceived risks on ex-ante consumption/savings decisions that is separate from the ex-post income impacts) has been challenging when conventional data sources are used, as these do not directly elicit ex-ante plans and perceptions at the individual level. This section shows that the coexistence of the same individual's specific perceived risks and consumption plan, as documented in the SCE, provides a rare opportunity to resolve this problem.[10] This contrasts with the common

[10] Guiso et al. (1992) was an early example of directly testing precautionary savings motives using the reported subjective income risks of Italian households. Other recent works that examine the impacts of expectations on readiness to spend include Bachmann et al. (2015) and Coibion et al. (2020). Recently, in closely related studies, Fuster et al. (2020) and Bunn et al. (2018) relied on survey answers to measure the stated marginal propensity to consume. Most related to this paper, Christelis et al. (2020) also found that

FIGURE 3. Perceived and realized job-separation and finding rates within groups



Note: Realized job-separation and finding rates are computed from the CPS, available at the Federal Reserve Bank of San Francisco's Labor Market Tracker. The sample is from January 2013 to December 2019. Both the realizations and the perceived probabilities are expressed over the quarterly horizon.

practice to date, which is to impute ex-ante unemployment risks to a particular individual from realized data based on only a few observable factors (Harmenberg and Öberg 2021).

I run a regression of the expected consumption growth reported in the SCE by each respondent on the same individual's expected wage growth, perceived wage risk, and unemployment risks under a range of specifications.

$$\mathbb{E}_{i,t}(\Delta c_{i,t+1}) = u_0 + u_1 \mathbb{E}_{i,t}(\Delta w_{i,t+1}) + u_2 \mathrm{Var}_{i,t}(\Delta w_{i,t+1}) + u_3 \mathrm{Prob}(U_{i,t+1}) + \eta_{i,t} \tag{14}$$

In the past, the literature operated on the assumption that such a reduced-form regression clearly corresponds to the commonly used Euler equation approximated to the second order (Parker and Preston 2005), where the expected consumption growth is equal to the sum of the intertemporal substitution and the precautionary savings motive. However, a linearly approximated Euler equation is reasonable only under a set of unrealistic and stringent assumptions, such as the absence of an external borrowing constraint, the absence of buffer-stock-savings behavior as elaborated in Carroll and Samwick (1997), and mild-sized income fluctuations, a point forcefully made by Carroll (2001) and Ludvigson and Paxson (2001). Therefore, in the regression results below, I primarily focus on testing the significance and qualitative effects of precautionary savings motives, without providing a structural interpretation of the size of the estimated coefficient.

expected consumption growth is positively correlated with perceived income risk at the individual level, based on Dutch households.

Across all of the specifications, as reported in Table 2, in addition to the significantly positive coefficient of the expected wage growth, which is consistent with buffer-stock-savings behavior, the perceived risk is positively correlated with the expected spending growth, as the precautionary savings motive predicts. Specifically, after controlling for individual fixed effects (e.g., the discount rate) and time fixed effects (e.g., the interest rate), each unit increase in the perceived variance is associated with around a 1.7 percentage point higher expected spending growth. Additionally, for the same individual, perceived unemployment risk, measured by the perceived probability of job separation in the next 12 months, is significantly and negatively correlated with expected consumption growth. This is sensible, as a higher perceived risk of unemployment not only increases the dispersion of future income outcomes but also lowers expected income. While controlling for perceived unemployment risk in Column (5) in Table 2, the coefficients of expected wage growth and perceived risk remain significant. This suggests that both wage risk and unemployment risk perceptions shape households' consumption plans, supporting this paper's approach of incorporating both. These estimates are correlations rather than causal effects, as the data provide no exogenous variation in perceived risk. The theory, however, does not require it: under the Euler equation, an optimality condition, a household's spending path and its risk perceptions are determined jointly, so that their co-movement is a property of the same decision problem. The fixed-effects specifications further restrict the relationship to within-person revisions rather than differences across households. Such revisions are ample enough to identify the coefficients: as reported in Section 4.3, individual fixed effects account for about two thirds of the variance in perceived risks, leaving the remaining third to vary within individuals over time.

TABLE 2. Perceived income risks and the household spending plan

	(1)	(2)	(3)	(4)	(5)
Expected wage growth	0.3236**	0.3062**	0.2537***	0.243***	0.1927***
	(0.122)	(0.1175)	(0.06118)	(0.06165)	(0.05071)
Perceived wage risk	6.127**	6.185**	2.096***	1.711**	2.006**
	(2.26)	(2.265)	(0.6034)	(0.6034)	(0.6299)
Perceived UE risk next 12m					-0.06348*
					(0.02623)
R-squared	0.000939	0.00318	0.953	0.953	0.959
Sample Size	56046	56046	56046	56046	51155
Time FE	No	Yes	No	Yes	Yes
Individual FE	No	No	Yes	Yes	Yes

Regression of expected spending growth on perceived income risks, corresponding to Equation 14. Standard errors, in parentheses, are clustered by household. *** p<0.001, ** p<0.01 and * p<0.05.

5. A life-cycle model with subjective risk perceptions

Section 4.6 provides reassuring evidence that individual consumption/savings decisions are indeed correlated with *their own* income expectations and perceived risks, regardless of the correctness of such perceptions. In this section, I show that recalibrating income

risks based on reported perceptions, in a standard incomplete-market life cycle model, also generates more empirically plausible predictions regarding inequality in liquid wealth compared to using indirect calibrations, and the difference is quantitatively important.

5.1. The baseline model

I reproduce a standard incomplete-market life-cycle model of consumption and saving featuring idiosyncratic income risks. All of the direct implications of perceived risks regarding liquid wealth accumulation and wealth inequality are sufficiently conveyed in the partial equilibrium. In Section B.4 in the Appendix, I consider the general equilibrium forces as in Huggett (1996), embedding a more realistic income risk profile and economic environment à la Carroll and Samwick (1997), Krueger et al. (2016), and Carroll et al. (2017).

In each period, a continuum of agents is born. Each agent i lives for L and works for T ($T \leq L$) periods after entering the labor market, during which they earn stochastic labor income y_τ at the working age of τ and face a constant survival probability $1 - D$. After retiring at the age of T, the agent lives for another $L - T$ periods and receives social security benefits. Without aggregate risks, there is no need to treat the calendar time t and the working age τ as two separate state variables; hence, I suppress the time subscript t from now on. All shocks are idiosyncratic.

5.1.1. Household's problem

Each period in life, the consumer chooses consumption to maximize their expected life-cycle utility under a discount factor β and a constant survival probability $1 - D$.

$$\max_{\{c_{i,\tau}\}_{\tau=1,2...L}} \mathbb{E}\left[\sum_{\tau=1}^{\tau=L}(1-D)^{\tau-1}\beta^{\tau-1}u(c_{i,\tau}) + (1-D)^{L-1}\beta^{L-1}\phi u(a_{i,L})\right] \tag{15}$$

where $c_{i,\tau}$ represents consumption at working age τ. The felicity function $u(c)$ takes a standard CRRA form with a relative risk aversion coefficient of ρ: $u(c) = \frac{c^{1-\rho}}{1-\rho}$. The second term is the homothetic bequest motive from the last period of life, derived from the post-consumption asset $a_{i,L}$, scaled by a strength parameter ϕ.

Denote the total cash in hand at the beginning of the period τ as $m_{i,\tau}$, the end-of-period savings in period τ after consumption as $a_{i,\tau}$, and the bank balance at the beginning of the period τ as $b_{i,\tau}$. Labor income y_τ is taxed at a rate of λ, and the social security tax rate is λ_{SS}. R is the gross real interest rate factor. The consumer starts with some positive bank balance in the first period of life, b_1, which may partly come from a lump-sum accidental bequest from the deceased population each period. The household makes consumption and savings decisions subject to the following intertemporal budget constraints.

$$\begin{aligned} a_{i,\tau} &= m_{i,\tau} - c_{i,\tau} \\ b_{i,\tau+1} &= a_{i,\tau}R \\ m_{i,\tau+1} &= b_{i,\tau+1} + (1-\lambda)(1-\lambda_{SS})\,y_{i,\tau+1} \\ a_{i,\tau} &\geq 0 \end{aligned} \tag{16}$$

The last inequality above is the no-borrowing constraint.

5.1.2. Income process

Each agent receives stochastic labor income between $\tau = 1$ and $\tau = T$ while they are of working age and receives a social security benefit after retirement. The income processes in both subperiods can be defined in a generic manner as described below. By allowing the possibility of persistent unemployment spells, the process is assumed to follow a slight variant of the standard permanent/transitory income process used in the literature (Carroll et al. 2017; Kaplan and Violante 2018). Specifically, $y_{i,\tau}$ is the product of the idiosyncratic wage rate[11] $w_{i,\tau}$ and the economy-wide wage rate W. The former consists of one permanent component $p_{i,\tau}$ and one potentially persistent or transitory component $\chi_{i,\tau}$. The aggregate wage is normalized to one; it becomes endogenous only in the general-equilibrium extension in Appendix B.4.

$$\begin{aligned} y_{i,\tau} &= e^{w_{i,\tau}} W \\ e^{w_{i,\tau}} &= e^{p_{i,\tau}} \chi_{i,\tau} \end{aligned} \tag{17}$$

During the working life, the permanent wage component is subject to a shock $\psi_{i,\tau}$ in each period and grows along a deterministic life-cycle profile governed by $\mathbb{G} = \{G_\tau\}_{\tau=1\ldots L}$.

$$e^{p_{i,\tau}} = G_\tau e^{p_{i,\tau-1}} e^{\psi_{i,\tau}} \tag{18}$$

The persistent/transitory shock $\chi_{i,\tau}$ takes different values depending on the employment status $\nu_{i,\tau}$, which is e when employed and u when unemployed.

$$\chi_{i,\tau} = \begin{cases} e^{\theta_{i,\tau}} & \text{if} \quad \nu_{i,\tau} = e \quad \& \quad \tau \leq T \\ \zeta & \text{if} \quad \nu_{i,\tau} = u \quad \& \quad \tau \leq T \\ \mathbb{S} & \text{if} \quad \tau > T \end{cases} \tag{19}$$

where ζ is the replacement ratio of the unemployment insurance and $\theta_{i,\tau}$ is the i.i.d. mean-zero shock to the transitory component of the wage conditional on staying employed.

Notice that this process also embodies the income process after retirement ($\tau > T$). The agent receives social security with a replacement ratio, $\mathbb{S}$, proportional to their permanent wage and the aggregate wage rate. That is, the effective pension benefit received is $\mathbb{S} e^{p_{i,\tau}} W$. I assume that the permanent component after retirement follows a deterministic path without additional stochastic shocks.

The parameters governing the degree of the income risk while the individual is of working age ($\tau \leq T$), in this model, consist of the standard deviations of the permanent and transitory wage shocks σ_ψ and σ_θ, respectively, as well as the transition probabilities of the job spells. For both types of wage shocks, I assume normal distributions, mean-normalized so that each shock has a unit mean in levels.

[11] This is equivalent to the usual interpretation of the wage rate in the literature as coming from idiosyncratic productivity under the implicit assumption of a perfectly inelastic labor supply.

$$\psi_{i,\tau} \sim N(-\frac{\sigma^2_{i,\psi}}{2}, \sigma^2_{i,\psi}) \tag{20}$$
$$\theta_{i,\tau} \sim N(-\frac{\sigma^2_{i,\theta}}{2}, \sigma^2_{i,\theta})$$

The transition matrix between unemployment ($\nu_{i,\tau} = u$) and employment ($\nu_{i,\tau} = e$) is the following (Krueger et al. 2016).

$$\pi(\nu_{i,\tau+1}|\nu_{i,\tau}) = \begin{bmatrix} \wp_i & 1-\wp_i \\ 1-E_i & E_i \end{bmatrix} \tag{21}$$

In general, this assumption implies to some degree that unemployment risks persist, but it conveniently nests the special case in which the unemployment risk is purely transitory when $\wp = 1 - E$, meaning the probability of unemployment is not dependent on the current employment status.

Crucially, note that all income risk parameters—σ_ψ, σ_θ, $\wp$, and E—carry a subscript i, indicating they may vary across individuals. The focus of this paper is to capture such heterogeneity, as reflected in the dispersion of perceived risks, rather than to follow the common practice of assuming homogeneous risks. This approach also avoids restricting heterogeneity in income risks to a single observable dimension, such as age.

5.1.3. Value function and consumption policy

The following two value functions characterize the consumer's problem in the last period of life ($\tau = L$) and all of the earlier periods ($\tau < L$), respectively.

$$V_\tau(\nu_{i,\tau}, m_{i,\tau}, p_{i,\tau}) = \max_{\{c_{i,\tau}, a_{i,\tau}\}} \quad u(c_{i,\tau}) + \phi u(a_{i,\tau}) \tag{22}$$

$$V_\tau(\nu_{i,\tau}, m_{i,\tau}, p_{i,\tau}) = \max_{\{c_{i,\tau}, a_{i,\tau}\}} \quad u(c_{i,\tau}) + (1-D)\beta \mathbb{E}_\tau \left[V_{\tau+1}(\nu_{i,\tau+1}, m_{i,\tau+1}, p_{i,\tau+1}) \right] \tag{23}$$

where the three state variables for the agents are the current employment status $\nu_{i,\tau}$, the total cash in hand $m_{i,\tau}$, and the permanent income $p_{i,\tau}$. $\nu_{i,\tau}$ drops out of the state variables in the special case of a purely transitory unemployment shock ($\wp = 1 - E$).

Under homogeneous risks, the solution to the problem above is a set of age-specific optimal consumption policies, $c^*_\tau(\nu_{i,\tau}, m_{i,\tau}, p_{i,\tau})$, and savings policies, $a^*_\tau(\nu_{i,\tau}, m_{i,\tau}, p_{i,\tau})$. Both are functions of all state variables. With ex-ante heterogeneity in income risks, such optimal behaviors further vary across potentially individual-specific perceived risks.

The population growth rate is n. With a deterministic life-cycle profile of survival probabilities, there exists a stable age distribution $\{\mu_\tau\}_{\tau=1,2,..L}$ such that $\mu_{\tau+1} = \frac{(1-D)}{1+n}\mu_\tau$ and $\sum_{\tau=1}^{L} \mu_\tau = 1$. The former condition reflects the probability of survival at each age and the latter is a normalization that guarantees that the fractions of all age groups sum

up to 1.[12]

5.1.4. Stationary distribution

Denote $x = \{m,\ p, \nu\} \in X$ as the idiosyncratic state of individuals. At any point in time, agents in the economy differ in age τ and idiosyncratic state x. The former is given by $\{\mu_\tau\}_{\tau=1,2,..L}$. For the latter, $\psi_\tau(B)$ represents the fraction of agents at age τ whose individual states lie in B as a proportion of all agents of age τ. The distribution of agents by age $\tau = 1$ depends on the initial condition of the labor income outcomes and the size of the accidental bequests, if any. For any other age $\tau = 2...L$, the distribution $\psi_\tau(B)$ evolves as the following.

$$\psi_\tau(B) = \int_{x \in X} P(x, \tau - 1, B) d\psi_{\tau-1} \quad \text{for all} \quad B \in B(X) \tag{24}$$

where $P(x, \tau - 1, B)$ is the probability that an agent will transit to B in the next period, conditional on the individual's state x at age $\tau - 1$. The transition function depends on the optimal consumption policy $c^*(x, \tau)$ at age τ and the exogenous transition probabilities of the income shocks.

5.2. The subjective model

The baseline model maintains the FIRE assumption, under which agents perceive exactly the calibrated risks. To capture the gap between perceived and calibrated risks identified in Section 4.4.1, I relax it: perceived risks now determine agents' consumption and savings decisions, while potentially differing from the true underlying risk parameters that govern the actual draw of income shocks. Denote the actual and perceived income risk parameters by $\Gamma = [\sigma_\psi, \sigma_\theta, \mho, E, \mathbb{G}]$ and $\tilde{\Gamma} = [\tilde{\sigma}_\psi, \tilde{\sigma}_\theta, \tilde{\mho}, \tilde{E}, \tilde{\mathbb{G}}]$, respectively. In various model specifications considered later, a subset of $\tilde{\Gamma}$ may differ from those in Γ. Furthermore, some of the parameters in $\tilde{\Gamma}$ or Γ may take different values across agents.

In any subjective model, it is $\tilde{\Gamma}$ that determines households' expectations of future income dynamics in their consumption/saving problems, as characterized in Equation 15 and value functions in Equation 22. Hence, the optimal consumption/saving $c^*_\tau(\nu_{i,\tau}, m_{i,\tau}, p_{i,\tau})$ and $a^*_\tau(\nu_{i,\tau}, m_{i,\tau}, p_{i,\tau})$ differ across agents with different $\tilde{\Gamma}$. In the meantime, the evolution of the asset distribution from age $\tau - 1$ to τ is governed by a new transition function $\tilde{P}$, which is different from the P in the baseline model defined in Equation 24. In contrast to the objective model, $\tilde{P}$ is a joint function of agents' optimal consumption and saving decisions, which are determined by their subjective beliefs about income risk, $\tilde{\Gamma}$, together with the objective income risk process, Γ, that governs the actual realization of income shocks and thus the resulting income distribution.

$$\tilde{\psi}_\tau(\tilde{B}) = \int_{\tilde{x} \in \tilde{X}} \tilde{P}(\tilde{x}, \tau - 1, \tilde{B}) d\tilde{\psi}_{\tau-1} \quad \text{for all} \quad \tilde{B} \in \tilde{B}(X) \tag{25}$$

Separating subjective from actual income processes also conceptually clarifies two distinct channels through which idiosyncratic income risk shapes asset accumulation and inequality. The first is the "choice" channel, which operates *ex-ante* via agents'

[12] With age-specific survival probability $1 - D_\tau$, the condition becomes $\mu_{\tau+1} = \frac{(1-D_{\tau+1})}{1+n}\mu_\tau \quad \forall \tau = 1, 2...L$, as discussed in Ríos-Rull (1996) and Huggett (1996).

consumption–saving decisions, driven by their *perceived* income risks. The second is the "outcome" channel, which operates *ex-post* through the *realized* dispersion of income shocks. To the extent that the subjective and actual income processes differ from each other, I can use the two as separate inputs to quantify the contribution of each channel.

5.3. Baseline calibration

This section discusses the calibration of the baseline model; Section 6.2 then discusses the variants of the subjective model in detail.

Calibrated income risks. Given the critical importance of the income risks assumption in my model, in addition to my estimates from the SIPP (as reported in Table 1), I thoroughly survey the risk estimates used in the existing incomplete-market macro literature, as summarized in Table A.4 in the Appendix.

Despite the disagreement in these estimates, the income risks used in these models are consistently larger than those reported in the survey. This is true even for the risk estimate most comparable to the surveyed PRs, the wage risk in Low et al. (2010). I use the median values of each parameter in the literature as the benchmark income risk profile, which is a combination of $\sigma_\psi = 0.15$ and $\sigma_\theta = 0.15$. These benchmark values are deliberately larger than my own SIPP-based estimates in Table 1 ($\hat{\sigma}_\psi \approx 0.10$ and $\hat{\sigma}_\theta < 0.03$ at annual frequency). Using the literature's values is therefore the conservative choice: higher calibrated risks generate more wealth inequality in the baseline model, leaving a smaller gap relative to the data for perceived risks to close. Following the calibration of Krueger et al. (2016), the yearly probability of staying unemployed is $\mho = 0.18$ and that of staying employed $E = 0.96$.

Life cycle. The model is set at the yearly frequency. The working age spans from 25 to 65 years old ($T = 40$), and the agent dies with certainty at age 85 ($L = 60$). The constant death probability before the terminal age is set as $D = 0.625\%$.

Regarding the deterministic permanent earning profile over the life cycle, $\mathbb{G}$, I draw on an age polynomial regression of the log total wage income from the SCF households aged 25-65. Using total wage income for the deterministic profile, rather than relying solely on wage rates from the primary job, naturally accounts for the deterministic profile of working hours over the life cycle. The estimated profile is very similar to those obtained by Gourinchas and Parker (2002), Cagetti (2003), and Kaplan and Violante (2014). The estimated growth profile is plotted in Appendix B.1. For the retirement phase, I assume a one-time drop of 50% in the permanent wage at age 66; that is, $G_{41} = 0.5$, and that the permanent wage stays flat until death. This produces an average expected growth factor of the permanent wage exactly equal to one over the entire working life. This serves as a normalization. Note that although alternative assumptions, such as a smoother decline of income after retirement, do change the wealth distribution across generations among the retired, they do not change the consumption/savings decisions because such a profile is entirely deterministic.

As to the public insurance policies, the unemployment insurance replacement ratio is set to the U.S. statutory level of $\zeta = 0.5$, following Krueger et al. (2016). (Appendix Table A.5 shows that under a much lower replacement ratio of 0.15, as in Carroll et al. (2017), the subjective model still significantly improves the model's empirical fit relative to the baseline.) The pension income relative to permanent income is assumed to be $\mathbb{S} = 65\%$. This, plus the 50% drop in permanent income, gives an effective deterministic wage drop of 70% from working age to retirement, which corresponds to an empirical

replacement ratio estimated for the U.S. economy. The unemployment insurance tax rate λ is set to balance the UI budget, while the social security tax rate λ_{SS} is fixed at the U.S. payroll tax rate (Table 3).

Initial conditions. Assumptions about the cross-sectional distribution of the initial permanent income and liquid asset holdings matter for the subsequent wealth inequality. I set the standard deviation of the log normally distributed initial permanent wage $p_{i,\tau}$ to 0.6 in order to match the heterogeneity in the wage income at age 25 from the SCF. The normalized bank balance at the beginning of a working life is set to 0.10, which is equal to the average liquid wealth-to-income ratio at age 25 in the SCF.

Preference. The coefficient of relative risk aversion ρ is 2. The discount factor β is set to be 0.96. I deliberately choose to fix the two preference parameters using consensus values instead of internally calibrating them to match the moments, such as the mean or median wealth/income ratios in the SCF.[13] I turn to the approach of indirectly inferring preference parameters to match data moments in Section 6.4.3.

Table 3 summarizes the parameters used in the calibration of the baseline model. This is nearly identical to what would be considered a standard calibration of an incomplete-market and liquid-assets model (Kaplan and Violante 2022).

TABLE 3. Parameters for the baseline model

Block	Parameter name	Values	Source
risk	σ_ψ	0.15	Median estimate from the literature
risk	σ_θ	0.15	Median estimate from the literature
risk	$U2U$	0.18	Median estimate from the literature
risk	$E2E$	0.96	Median estimate from the literature
initial condition	$\sigma_\psi^{\text{init}}$	0.629	Estimated for age 25 in 2016 SCF
initial condition	b_1	0.106	Average liquid wealth to income ratio at age 25 in 2016 SCF
life cycle	n	0.005	U.S. census
life cycle	T	40	Standard assumption
life cycle	L	60	Standard assumption
life cycle	$1-D$	0.994	Standard assumption
life cycle	$\mathbb{G}$	Figure A.8A	Estimated using wage inc. in 2016 SCF
prices	R	1.01	Standard calibration
prices	W	1	Normalization
preference	ρ	2	Standard calibration
preference	β	0.96	Standard calibration
bequest	ϕ	1.0	Standard calibration
policy	$\mathbb{S}$	0.65	U.S. average
policy	λ	N/A	Balances the UI budget
policy	λ_{SS}	0.065	U.S. social security payroll tax
policy	ζ	0.5	U.S. statutory replacement rate

Note: parameters used in the baseline model. All parameters, whenever relevant, are at the annual frequency.

6. Model results

[13] Kaplan and Violante (2022) discusses in detail how internally calibrated discount factors in one-asset models differ depending on whether they are targeting liquid wealth or total net worth. Their calibration of β is 0.945 for a targeted liquid-asset-to-income ratio of 0.6 and 0.98 for a targeted net-worth-to-income ratio of 4.6. This is the same as the average value estimated in models with heterogeneous time preferences, as in Carroll et al. (2017) and Krueger et al. (2016).

6.1. Baseline model with calibrated income risks

I first examine the patterns of wealth accumulation and inequality generated from a benchmark calibration, as reported above. In particular, under a set of standard parameterizations on permanent and transitory wage risks at annual frequencies of $\sigma_\psi = 0.15$ and $\sigma_\theta = 0.15$, and unemployment risks of $U2U = 0.18$ and $E2E = 0.96$, the baseline of Figure 5 reproduces the well-known result (for surveys, see Guvenen 2011; De Nardi 2015; Kaplan and Violante 2018) that a carefully calibrated standard one-asset incomplete-market model without additional heterogeneity, such as that in time discount rates, predicts significantly less wealth inequality (a Gini coefficient of 0.73) than the liquid wealth inequality observed in the data. The distribution of net liquid wealth in the 2016 SCF based on the definition of Kaplan et al. (2014) and Carroll et al. (2017)[14] has a Gini coefficient of 0.89 (negative liquid wealth excluded to be consistent with the model).

The second major discrepancy between the model and the data is that the model substantially underdelivers the fraction of agents who are close to their borrowing constraints. Specifically, the baseline model implies that only 9% of the population is hand-to-mouth (H2M)—defined as agents whose liquid wealth does not exceed their half-month income—whereas the corresponding share based on net liquid wealth in the SCF is 0.35. The baseline model also significantly underpredicts the H2M shares at virtually all ages compared to what is observed empirically (Figure 5). In the SCF, the H2M share starts at around 40% between ages 25 and 45, then gradually falls to below 10%, with a brief increase immediately after retirement. The discrepancy is consistent with the fact that, in this model, strong precautionary saving motives make agents build asset buffers and remain far from their borrowing limits. As a result, even the young agents who start off their working life mostly constrained are able to leave H2M status shortly afterward.

In addition to wealth moments, I can also compare the model-implied marginal propensities to consume (MPCs) with those elicited in the SCE. This follows the use of survey-based stated MPCs by Christelis et al. (2019) and Fuster et al. (2020). Recognizing that the MPCs from the SCE are elicited as the fraction of a hypothetical $500 windfall to be consumed, I compute the model-based MPCs using the same definition.[15] Because the model generates relatively few H2M households, the average MPC in the stationary distribution is only 0.17, about 9 percentage points below the SCE average of 0.26. Unlike the relatively flat empirical age profile, which remains in the 0.2–0.35 range, the model produces a pronounced U-shaped MPC pattern over the life cycle. The differences in optimal consumption/saving policies and shifts of the asset distribution across ages jointly drive this pattern. On one hand, in standard life-cycle settings, the optimal consumption rule implies that, for a given level of wealth, younger agents have higher MPCs than older ones. On the other hand, young agents in the model are disproportionately concentrated at low wealth levels, which raises their average MPC.

Despite these differences, the baseline model still generates a hump-shaped profile of average wealth over the life cycle, closely matching the net liquid wealth pattern

[14] According to this definition, liquid assets include checking, savings, money-market funds, government bonds, directly held mutual funds, stocks and corporate bonds; and liquid debt is the sum of all credit card balances that accrue interest after the most recent payment.

[15] Unlike the theoretical definition of MPC as the partial derivative of the consumption function with respect to market resources, $\frac{\partial}{\partial m_{i,\tau}} c^*_\tau(\nu_{i,\tau}, m_{i,\tau}, p_{i,\tau})$, the MPC calculated here, $\frac{c^*_\tau(\nu_{i,\tau}, m_{i,\tau}+\Delta m, p_{i,\tau}) - c^*_\tau(\nu_{i,\tau}, m_{i,\tau}, p_{i,\tau})}{\Delta m}$, varies with the size of Δm.

observed in the SCF (Figure 5). Allowing for voluntary bequests in the final period prevents retired agents from fully running down their assets. In Appendix Figure A.10, I also show that the model-implied expected consumption growth follows an intuitive life-cycle pattern, gradually declining from youth and flattening at older ages. Echoing the finding in Table 2, the ex-ante expectations data display patterns that align closely with the model's predictions.

6.2. Subjective model calibrations

The preferred specification/calibration in this paper is a fully specified subjective model as defined in Section 5.2, referred to as the "Subjective" model. In this model, the actual wage and unemployment risks are kept almost the same as the baseline and remain homogeneous (with a minor adjustment of wage risk due to unobserved heterogeneity). Meanwhile, the level and the heterogeneity in perceived wage and unemployment risks are recalibrated to be consistent with the survey data. The recalibration takes into account all the stylized patterns discussed in the first part of the paper: (a) perceived wage risks are lower on average than the true wage risk, stemming from both unobserved heterogeneity and overconfidence; (b) perceived wage risks are heterogeneous across agents; and (c) perceived unemployment risks are heterogeneous. Below, I describe these three incremental recalibrations in detail. The exact parameter values are reported in Table 4, together with the other alternative specifications considered later on.

Lower perceived wage risks and heterogeneous growth rates. Recalibrating perceived wage risks to a lower level requires disciplining the two sources of the wedge identified in Section 4.4.1, using the estimates of σ_ξ and δ. On the perceived side, $\tilde{\sigma}_\psi$ and $\tilde{\sigma}_\theta$ are set to their estimated values of 0.034 and 0.021, respectively (implying a permanent-transitory ratio of $\kappa = 1.62$), which together match the total PR observed in the SCE,[16] and the perceived life-cycle growth profiles $\tilde{\mathbb{G}}$ are heterogeneous with a dispersion of $\sigma_{\xi,\psi} = 0.028$. On the actual side, the true risks σ_ψ and σ_θ are 0.147, and the actual growth profiles $\mathbb{G}$ are the same as perceptions, since the identification results attribute the expectational dispersion entirely to unobserved heterogeneity. Combined, this is the "Subj. Wage (Uniform)" calibration in Table 4. Since it bundles the lower perceptions with the heterogeneous growth profiles, Table 4 also reports "Subj. Wage (Lower)", which lowers the perceptions alone. Section 6.3 uses the pair to separate the two.

Only the permanent part of σ_ξ enters these growth profiles. I assume that half of the 4-percentage-point heterogeneity identified in Section 4.4.1 reflects the permanent difference in wage growth rates, that is $\sigma_{\xi,\psi} = \sqrt{0.04^2/2} = 0.028$. The remaining half is transitory, $\sigma_{\xi,\theta} = 0.028$; I abstract from it here, as its effects largely average out over the life cycle. This simplification likely understates unobserved heterogeneity, making the resulting calibration a conservative lower bound. I assign each agent to one of three equally likely deterministic wage growth profiles, designed to match an annual standard deviation of the same size.[17] Figure A.9 visualizes these profiles, with the mean

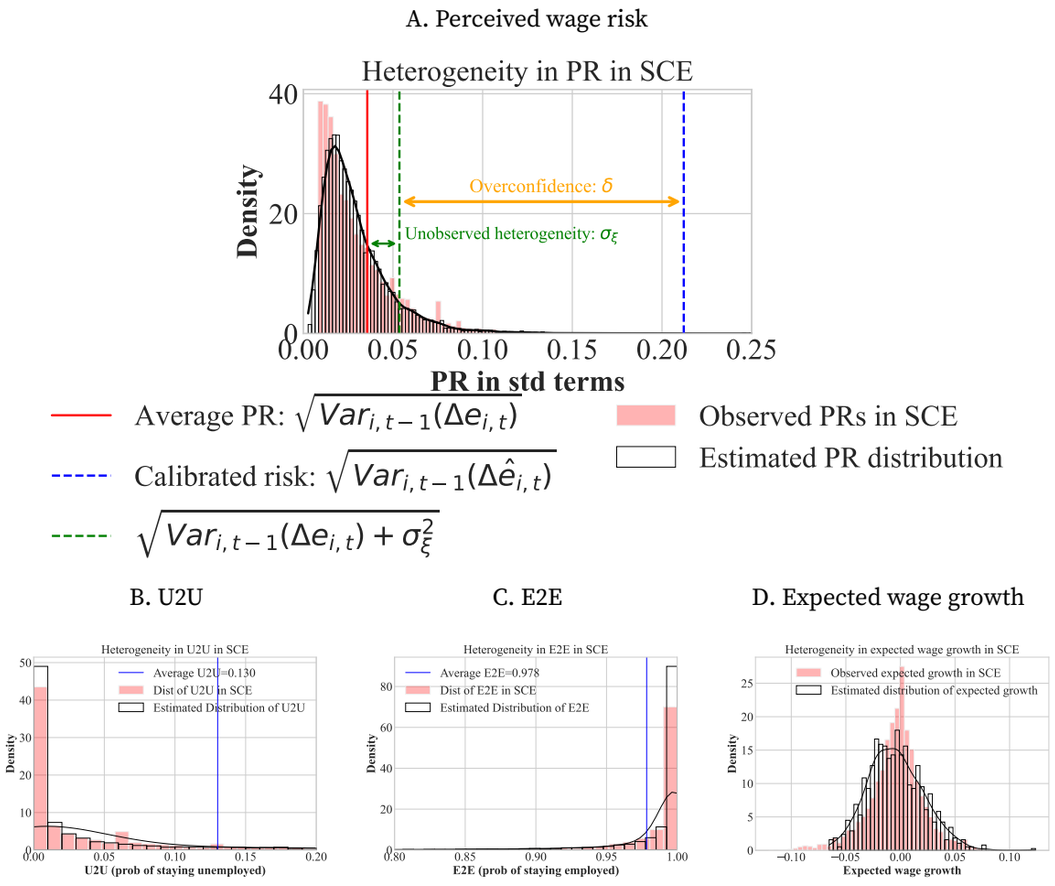
[16] In Appendix Table A.5, I show that under alternative ratios of permanent and transitory component $\kappa = \frac{\tilde{\sigma}_\psi}{\tilde{\sigma}_\theta}$, the subjective model remains a significant improvement relative to the baseline in terms of the empirical fit.

[17] With the wage growth expectations across agents approximately following a normal distribution with mean μ and variance σ^2, the three equally probable growth rates can be approximated by $\mu - \sqrt{3/2}\sigma$, μ, and $\mu + \sqrt{3/2}\sigma$, respectively. Appendix Table A.5 reports the model under five equiprobable profiles

trajectory corresponding to the baseline calibration $\mathbb{G} = \{G_\tau\}_{\tau=1,\ldots,L}$.

Heterogeneous perceived wage risks. The second recalibration is more straightforward: on top of the above, agents perceive heterogeneous wage risks, corresponding to the "Subj. Wage" specification reported in Table 4. This means that $\tilde{\sigma}_\psi$ and $\tilde{\sigma}_\theta$ take different values across agents. I directly calibrate the heterogeneity in PRs by fitting a truncated log-normal distribution to the cross-sectional distribution of the time-average PRs in the SCE, as shown in Figure 4A. The estimated distribution is then discretized into equally probable values used in the model, i.e., $\{0.01, 0.02, 0.04\}$ for both $\tilde{\sigma}_\psi$ and $\tilde{\sigma}_\theta$. The figure also visually illustrates the difference between the average perceived risk and calibrated risk stemming from the combination of unobserved heterogeneity σ_ξ and overconfidence δ, both of which are separately identified.

FIGURE 4. Calibration of heterogeneous perceived risks in the SCE



Note: The top figure plots the observed distribution of the perceived income risks from the SCE and the fitted truncated log-normal distribution. The exact procedure of identifying σ_ξ and δ is discussed in Section 4.4.1. The other three figures plot the distribution of unemployment risks and expected wage growth in the survey, as well as their fitted distribution.

Heterogeneous perceived unemployment risks. Third, in addition to subjective wage risks, agents also perceive heterogeneous unemployment risks, meaning that $\tilde{E}$ and $\tilde{U}$ differ across agents. Stacking this on top of the two wage-risk recalibrations above yields the full subjective model, labeled "Subjective" in Table 4. The heterogeneity in the perceived unemployment risk is calibrated similarly by fitting a truncated log-normal distribution to the cross-section of perceived $U2U$ and $E2E$ probabilities (Figure 4).[18] The estimated distribution is further discretized into three equally prob-

and under Gauss-Hermite quadrature with three and five nodes; the key moments move by only one to two percentage points.

[18] Informed by the observed low correlation at the individual level between perceived wage risks, job separation, and finding probabilities in the SCE, I estimate the distribution of U2U and E2E independently

able grid points [0.0, 0.006, 0.321] for $U2U$ and [0.96, 0.998, 1.0] for $E2E$. According to these profiles, approximately one-third of the agents in the economy face no risks of persistent unemployment spells, either through high job-finding rates or nearly zero job-separation rates. Meanwhile, one-third of the agents face potentially long durations of unemployment with lower expected incomes and higher probabilities of hitting their borrowing constraints.[19]

6.3. Results and anatomy of the subjective model

Results from the subjective model. The recalibrated model, by incorporating subjective perceptions of risk, markedly enhances its fit to the data across a range of moments. Figure 5 displays key moments of the stationary distribution generated by the subjective model alongside those of the baseline. The subjective model pushes the Lorenz curve further outward, yielding an 11-percentage-point rise in the Gini coefficient—from 0.73 in the baseline to 0.84. In the subjective model, the share of hand-to-mouth (H2M) agents is 31%, which is 22 percentage points higher than in the baseline, indicating that a large portion of households are close to liquidity constraints. A particularly striking feature is that the subjective model produces a life-cycle pattern of H2M incidence that closely mirrors the empirical data. As a result, the model-implied aggregate MPC increases from 0.17 to 0.25, driven mainly by higher MPCs among young and middle-aged households. The model still delivers a pronounced decline in MPCs over working life as assets are accumulated, in contrast with the nearly flat empirical MPC profile from youth through old age. Matching the life-cycle pattern of MPCs is beyond the ambition of this paper. A stable and flat MPC profile over life is likely due to reasons unrelated to expectations and observed demographic variables, and instead to the presence of substantial, latent heterogeneity, as shown by Lewis et al. (2024); Kosar and Melcangi (2025).

The rest of this section breaks down the improvement of the subjective model relative to the baseline into different mechanisms.

Wage risks versus unemployment risks. The recalibration of wage risks and heterogeneous perceived unemployment risks contribute to the improved model fit, but to varying degrees. These correspond to the "Subj. Wage" and "Subj. UE" specifications in Table 4, whose rows in Table 5 show the model moments from each of these specifications.

First, the recalibration of wage risks is the primary driver of the higher liquid wealth inequality. Introducing it alone increases the Gini coefficient by 8 percentage points, while recalibrating unemployment risks alone adds only 2; jointly, they deliver the 11-percentage-point increase of the subjective model. Second, the two channels are strongly complementary in generating hand-to-mouth households. Wage risk recalibration alone raises the H2M share from 9% to 18%, and unemployment risks alone to 15%, but jointly they raise it to 31%—exceeding the sum of the two separate contributions. Intuitively, with unemployment benefits at the statutory replacement rate, heterogeneous unemployment risk bites mostly for agents who also perceive low wage risks and thus hold little precautionary wealth; both forces together particularly expand the mass of

from that of the wage risks. The correlation between perceived risks and U2U and E2E probabilities is 0.07, and -0.1, respectively; the correlation between job separation and job finding expectations is 0.01.

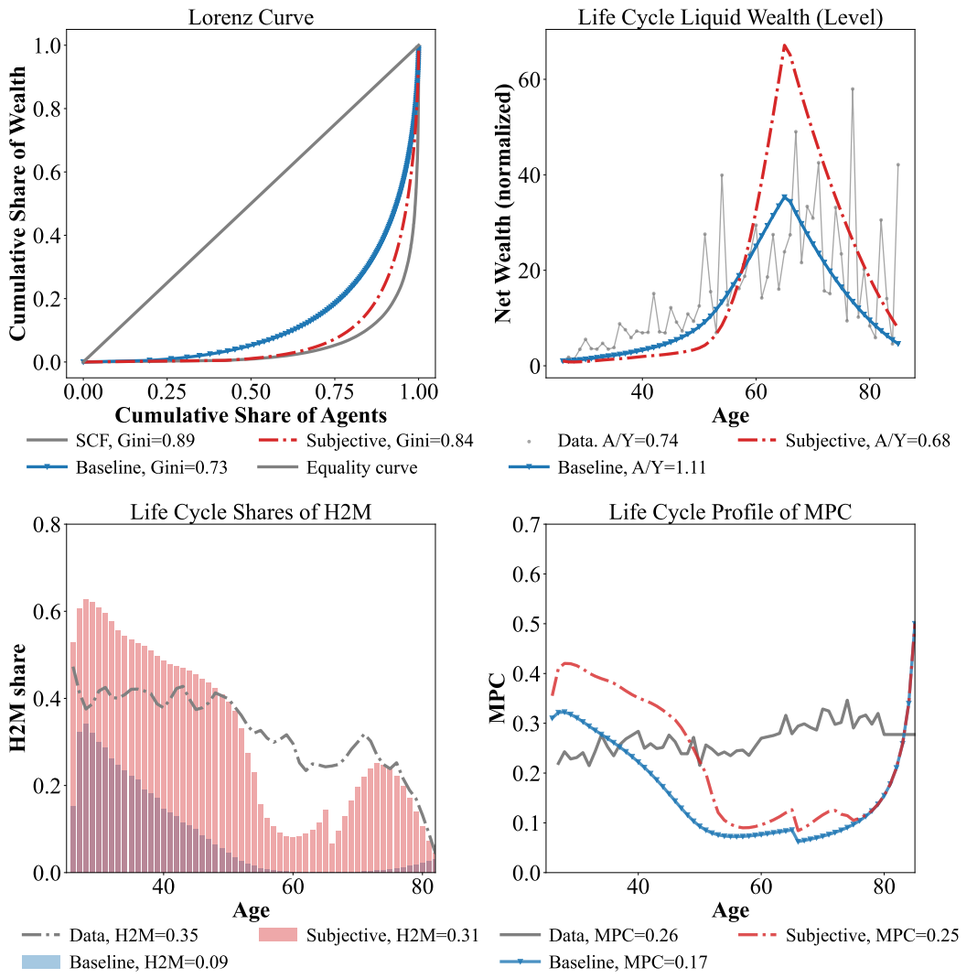
[19] See Ahn and Hamilton (2020), Ozkan et al. (2023), Gregory et al. (2025), and Castro et al. (2025) for evidence on ex-ante heterogeneity in job transitions; Savoia (2023) and Oh and Rogantini Picco (2025) for the macroeconomic consequences of such heterogeneity.

TABLE 4. Model parameters: perceived versus actual

	Perceived					Actual				
Model	$\tilde{\sigma}_\psi$	$\tilde{\sigma}_\theta$	$\tilde{E}$	$\tilde{\upsilon}$	$\tilde{\mathbb{G}}$	σ_ψ	σ_θ	E	υ	$\mathbb{G}$
Baseline	0.15	0.15	0.96	0.18	Fig. A.8A	0.15	0.15	0.96	0.18	Fig. A.8A
Subj. Wage (Lower)	0.034	0.021	0.96	0.18	Fig. A.8A	0.15	0.15	0.96	0.18	Fig. A.8A
Subj. Wage (Uniform)	0.034	0.021	0.96	0.18	Fig. A.8B	0.147	0.147	0.96	0.18	Fig. A.8B
Subj. Wage	{ 0.01, 0.02 0.04 }	{ 0.01, 0.02 0.04}	0.96	0.18	Fig. A.8B	0.147	0.147	0.96	0.18	Fig. A.8B
Subj. UE	0.15	0.15	{ 0.96, 0.998, 1.0 }	{ 0.0, 0.006 0.321}	Fig. A.8A	0.15	0.15	0.96	0.18	Fig. A.8A
Subjective	{ 0.01, 0.02 0.04}	{ 0.01, 0.02 0.04}	{ 0.96, 0.998, 1.0 }	{ 0.0, 0.006 0.321}	Fig. A.8B	0.147	0.147	0.96	0.18	Fig. A.8B
Subj. Wage + Obj. Het. UE	{ 0.01, 0.02 0.04}	{ 0.01, 0.02 0.04}	{ 0.96, 0.998, 1.0}	{ 0.0, 0.006 0.321}	Fig. A.8B	0.147	0.147	{0.96, 0.998, 1.0 }	{ 0.0, 0.006 0.321}	Fig. A.8B
Obj. Het. Wage + Subj. UE	{ 0.01, 0.02 0.04}	{ 0.01, 0.02 0.04}	{ 0.96, 0.998, 1.0}	{ 0.0, 0.006 0.321}	Fig. A.8B	{ 0.01, 0.02 0.04}	{ 0.01, 0.02 0.04}	0.96	0.18	Fig. A.8B
Obj. Het. Both	{ 0.01, 0.02 0.04 }	{ 0.01, 0.02 0.04}	{0.96, 0.998, 1.0 }	{ 0.0, 0.006 0.321}	Fig. A.8B	{ 0.01, 0.02 0.04}	{ 0.01, 0.02 0.04 }	{0.96, 0.998, 1.0}	{ 0.0, 0.006 0.321}	Fig. A.8B
Biased Exp.	{ 0.01, 0.02 0.04}	{ 0.01, 0.02 0.04}	{0.96, 0.998, 1.0}	{ 0.0, 0.006 0.321}	Fig. A.8B	0.147	0.147	0.96	0.18	Fig. A.9 (d)
Fanning	{ 0.01, 0.02 0.04}	{ 0.01, 0.02 0.04 }	{0.96, 0.998, 1.0}	{ 0.0, 0.006 0.321}	Fig. A.9 (a)	0.147	0.147	0.96	0.18	Fig. A.8B
Hump	{ 0.01, 0.02 0.04}	{ 0.01, 0.02 0.04}	{0.96, 0.998, 1.0}	{ 0.0, 0.006 0.321}	Fig. A.9 (b)	0.147	0.147	0.96	0.18	Fig. A.8B
Right skew	{ 0.01, 0.02 0.04}	{ 0.01, 0.02 0.04}	{0.96, 0.998, 1.0}	{ 0.0, 0.006 0.321}	Fig. A.9 (c)	0.147	0.147	0.96	0.18	Fig. A.8B

Note: The table shows all parameterizations under different model assumptions. Heterogeneity in any parameter is directly expressed as equally probable values, discretized from its estimated underlying distribution.

FIGURE 5. Key model moments: baseline versus subjective



Note: The figure displays model-implied moments alongside their empirical counterparts, for both the baseline and the subjective model. Shown are: the Lorenz curve of household wealth (upper left); life-cycle profiles of average wealth normalized by its value at age 25 (upper right); the share of H2M households over the life cycle (bottom left); and the life-cycle profile of MPCs, measured as the reported marginal propensity to consume out of a hypothetical $500 windfall (bottom right). All wealth statistics are based on net liquid wealth from the 2016 SCF, and reported MPCs are obtained from the SCE.

borrowing-constrained agents at the left end of the distribution.

At the same time, wage and unemployment risk recalibration contribute 4 and 2 percentage points on their own, respectively, to a total 8-point increase in the average MPC. While the model-implied MPCs are tightly linked to the wealth distribution, perceived risks also directly shape ex-ante decisions conditional on wealth. Therefore, a higher model-implied MPC does not entirely channel through a better prediction of the wealth distribution. This suggests that recalibrating both wage and unemployment risks is crucial for improving the model's predictions of MPCs.

Choice versus outcome. I can also examine whether the subjective model's better fit to data patterns comes from the "choice" or the "outcome" channel (Section 5.2), or exactly from the wedge between perceived risks and actual income risks. I find that the "choice" channel is the key. To put it differently, what matters is the recalibration of perceived risks according to the survey, regardless of whether such perceptions are also the actual income processes. In the three variants of the subjective model, where either the actual unemployment risks or wage risks are assumed to be the same as perceptions, the wealth Gini, the H2M share, and average MPCs are rather similar to

those of the subjective model. In the first version, the actual $E = \tilde{E}$ and $\mho = \tilde{\mho}$, taking subjective heterogeneous values. In the second version, $\sigma_\psi = \tilde{\sigma}_\psi$ and $\sigma_\theta = \tilde{\sigma}_\theta$, both taking subjective values. In the third version, both actual wage risks and unemployment risks align with perceptions. Their statistics are reported under "Subj. Wage + Obj. Het. UE", "Obj. Het. Wage + Subj. UE", and "Obj. Het. Both" in Table 5.

Overconfidence versus heterogeneity. The full subjective model incorporates both ex-ante heterogeneity in life-cycle growth rates and overconfidence in that perceived wage risks are significantly lower than the true risks. It is interesting to ask which of the two channels contributes to the improved model fit. The "Subj. Wage (Lower)" row of Table 5 isolates overconfidence on its own: it is identical to the baseline in every actual respect, and differs only in that agents perceive the lower, survey-implied risks. The step from "Baseline" to "Subj. Wage (Lower)" therefore measures overconfidence alone, and the further step to "Subj. Wage (Uniform)" adds the heterogeneity in growth profiles. The two channels turn out to divide the labor. Heterogeneity in growth profiles accounts for nearly all of the improvement in wealth inequality, raising the Gini by 8 percentage points against 1 for overconfidence. Overconfidence, in contrast, is what brings households to their borrowing constraints: on its own it raises the H2M share by 7 percentage points and lowers the wealth-to-income ratio from 1.11 to 0.90, while growth heterogeneity adds a further 2 percentage points of H2M. Intuitively, dispersion in permanent income growth widens the distribution of wealth without moving many agents to its lower end, whereas perceiving less risk than one actually faces erodes precautionary buffers for everyone. These two channels are precisely the two sources of the perception wedge identified in Section 4.4.1; they improve the model along different dimensions, and both are important.

To summarize, the subjective model highlights the central finding of this paper: even though the perceived income risks reported in surveys may not perfectly correspond to the objective processes that govern stochastic income shocks, households make savings decisions based on these perceptions. As a result, once the perceptions are recalibrated to reflect the level and heterogeneity in the survey, the model generates predictions about wealth accumulation that are more consistent with the empirical data. Further breakdown suggests that, within this improvement in model predictions, incorporating realistic heterogeneity in perceived risks is particularly important. This incremental improvement of the subjective model over the baseline is robust to alternative baseline calibrations—including a lower level of calibrated income risks, alternative unemployment-insurance replacement ratios, and alternative splits between the perceived permanent and transitory components of wage risk—as reported in Appendix Table A.5. The following sections assess the additional robustness of this claim across alternative specifications and assumptions.

6.4. Additional considerations

6.4.1. Biases in expectations

In this section, I consider the importance of biases in the expected growth rate. Recall that in Section 4.4.1, I caution that the expectational dispersion across agents may not fully reflect unobserved heterogeneity σ_ξ, and may instead partly reflect expectational biases $\varepsilon_{i,t}$. This essentially implies that the agents subjectively view their life cycle profiles as more distinct than the true heterogeneity. One mechanism through which

TABLE 5. Summary of model results and data

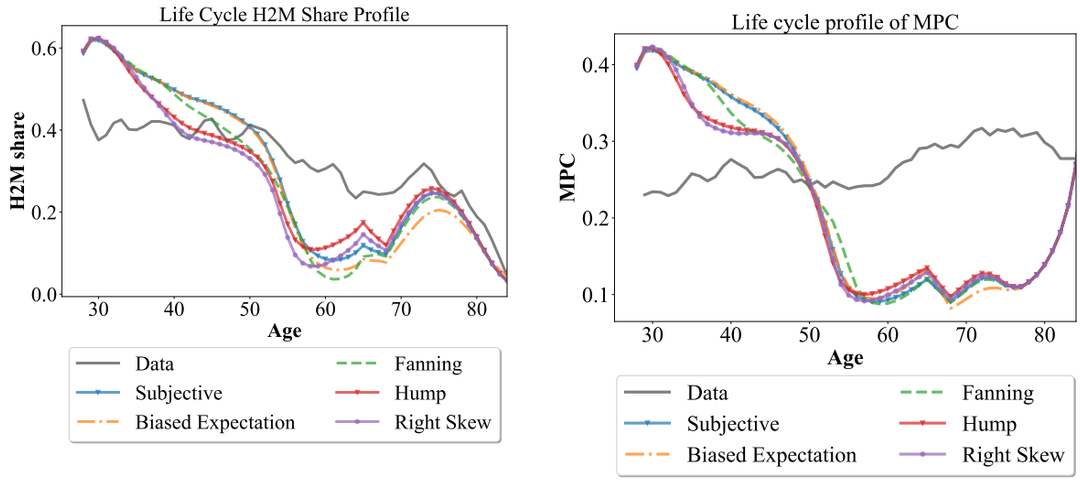
Model/Data	Gini	Bottom 0.9	Bottom 0.7	Bottom 0.5	Average A/Y	H2M share	Average MPC
Data	0.89	0.15	0.04	0.01	0.74	0.35	0.26
Baseline	0.73	0.40	0.14	0.05	1.11	0.09	0.17
Subj. Wage (Lower)	0.74	0.39	0.13	0.04	0.90	0.16	0.20
Subj. Wage (Uniform)	0.82	0.26	0.07	0.02	0.82	0.18	0.21
Subj. Wage	0.81	0.29	0.08	0.02	0.78	0.18	0.21
Subj. UE	0.75	0.37	0.12	0.04	1.00	0.15	0.19
Subjective	0.84	0.25	0.05	0.01	0.68	0.31	0.25
Subj. Wage + Obj. Het. UE	0.84	0.25	0.05	0.01	0.72	0.29	0.24
Obj. Het. Wage + Subj. UE	0.83	0.28	0.04	0.01	0.60	0.33	0.26
Obj. Het. Both	0.83	0.29	0.05	0.01	0.63	0.32	0.25
Biased Expectation	0.80	0.32	0.08	0.02	0.59	0.30	0.25
Fanning	0.83	0.26	0.06	0.01	0.67	0.31	0.24
Hump	0.84	0.26	0.06	0.01	0.74	0.31	0.24
Right Skew	0.83	0.27	0.06	0.01	0.76	0.30	0.24
CRRA	0.87	0.20	0.03	0.00	0.94	0.34	0.27
Patience	0.86	0.23	0.04	0.01	0.86	0.32	0.25
Bequests	0.85	0.24	0.05	0.01	0.73	0.33	0.25
Borrowing	0.89	0.18	-0.02	-0.05	0.59	0.47	0.22

Note: The table shows the model-implied Gini coefficients, the wealth shares owned by the bottom 90, 70, and 50 percent of agents, the mean wealth-to-income ratio, the shares of hand-to-mouth agents (H2M), and average MPC in the partial-equilibrium stationary distribution under various model specifications. H2M is defined as those whose liquid wealth is no more than two weeks' (1/24 of annual) income. MPC is calculated in the model as the proportion of a hypothetical one-time $500 windfall spent, as measured in the SCE. Other wealth statistics in the data are computed for net liquid wealth in the 2016 SCF. The specifications build up one channel at a time. Relative to the baseline, "Subj. Wage (Lower)" lowers the *level* of perceived wage risk; "Subj. Wage (Uniform)" adds heterogeneous growth profiles; "Subj. Wage" adds heterogeneous perceived wage risks; and the preferred "Subjective" model adds heterogeneous perceived unemployment risks. "Subj. UE" instead adds only the latter to the baseline. The three "Obj. Het." rows start from the subjective model and set the actual unemployment risks, wage risks, or both equal to perceptions. The remaining rows modify the subjective model's growth profiles, preferences, or borrowing constraint. See Table 4 for the exact parameterization of each specification.

such patterns emerge might be the "overpersistence" bias as argued by Rozsypal and Schlafmann (2023): if agents systematically perceive their transitory shocks to be more persistent than they are, there will be biases in income expectations leading to a wider expectational dispersion than the actual income realizations. Since I cannot directly identify the size of the biases, but only infer their upper bound, I treat this as a robustness check rather than my preferred specification. In particular, I recalibrate the subjective model by letting subjective life-cycle profiles $\tilde{\mathbb{G}}$ be different from the objective ones. I assume that only half of the observed dispersion in expected wage growth rates in $\sigma_\xi = 0.04$ is due to true unobserved heterogeneity, whereas the other half stems from idiosyncratic biases in their respective expectations. This effectively makes $\sigma_{\xi,\psi}$, the dispersion in permanent growth rate, about half of the previous calibration. The exact parameterization is reported in Table 4. Under this assumption, agents expect more dispersion in growth rates than the actual heterogeneity. The dispersion of the true life cycle growth profiles shrinks, as shown in Figure A.9.

Figure 6 plots the results under such assumptions in addition to the preferred specification. This variant yields a lower Gini of liquid wealth than the preferred model, which is nevertheless still 7 percentage points higher than the baseline. This is because limiting the true heterogeneity in life cycle wage growth profiles also limits the ultimate income inequality. However, the fraction of H2M only shrinks slightly from 31% to 30%, which is still substantially larger than the baseline's prediction. The model-implied MPCs remain at 0.25, as in the subjective model. Taking this evidence together, the model's improved fit to data is robust to the existence of expectational biases in wage growth rate expectations.

FIGURE 6. Key model moments from the subjective model: other considerations



Note: The figure shows, under various alternative model assumptions, the life-cycle profiles of the H2M share (left) and the average MPC (right), compared to their data counterparts from the 2016 *Survey of Consumer Finances* (SCF) and the *Survey of Consumer Expectations* (SCE), respectively. All profiles are plotted as three-year moving averages.

6.4.2. Unfolding of expectational heterogeneity over the life cycle

With heterogeneous life-cycle profiles of wage growth rates, which are partially or fully expected by the agents, I assume that the agents perceive that the degree of permanent heterogeneity in growth rate per year over the life cycle stays the same. This implies that the heterogeneity in the level of the wage accumulates at a constant pace over the working life before retirement. Due to a lack of data on the entire life-cycle expectations of wage rates, this comes as a natural assumption for my baseline. However, it may also be plausible to assume that the predictable heterogeneity across agents differs over the stages of working life. It is also possible that such heterogeneity is not fully reflected in agents' expectations early on and gradually becomes so (Guvenen 2007).

Assuming the actual deterministic profiles $\mathbb{G}$ stay the same as in the subjective model, I consider three alternative subjective unfolding patterns of life-cycle wage heterogeneity. All profiles share the same average yearly heterogeneity in wage growth rates of 2.8 percentage points as $\sigma_{\xi,\psi}$, but differ in their relative shapes over the life cycle. The first is a "fanning-out" profile in that the perceived heterogeneity in growth rates monotonically increases over the life cycle. As life unfolds, agents increasingly become dispersed in their expected wage growth rates. The second is hump-shaped, in that the heterogeneity widens in the middle of working life, then shrinks later. The third has a right-skewed unfolding shape in that the heterogeneity starts from the highest level and gradually dissipates over the life cycle. This dynamic may have to do with learning over the life cycle, whereby agents gradually accumulate observations to correctly infer their true permanent income.

The results under these variants of the preferred model are plotted in Figure 6 and reported in Table 5. In terms of asset Gini, H2M share, and MPC, different models yield almost identical results. The major differences between these models lie in their steady-state level of the liquid wealth to income ratio. With either a hump-shaped or right-skewed profile, the wealth-to-income ratio is slightly higher than in the other cases. All in all, the takeaway is that the subjective model's overall improvement is

also robust to different unfolding patterns of life-cycle heterogeneity, as long as such heterogeneity is already incorporated.

6.4.3. Calibration via indirect inference

My baseline calibration, as well as the recalibration, externally fixes the preference parameters of the model. The purpose there is to show the improvement in model fit due to calibrating perceived income risks without altering preference parameters. In this section, I pursue an alternative strategy, which has been the convention of the existing literature. That is, the preference parameters are internally calibrated to match observed moments. The purpose here is to show that incorporating subjective perceptions of income risks also improves the model's fit across different moments.

For both the baseline model and the preferred subjective model, I consider separately targeting the following seven data moments that are often of interest in the literature. The first four are life cycle profiles of (a) liquid wealth, (b) H2M shares, (c) MPCs and (d) the expected consumption growth factors $\tilde{\mathbb{E}}_{i,\tau}(\frac{c_{i,\tau+1}}{c_{i,\tau}})$. In addition to targeting life cycle profiles, I also target three population average moments directly, namely the average (e) H2M ratio, (f) MPC, and (g) liquid wealth to income ratio. The expected consumption growth profile is from the SCE.

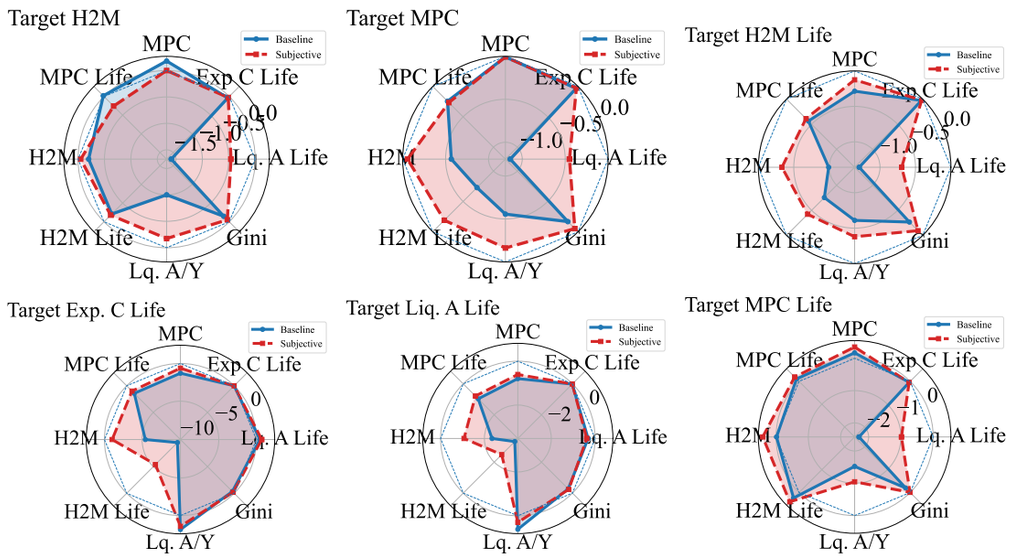
Among these calibration strategies based on different targeted moments, what is particularly novel is the one that targets the *expected* consumption growth, rather than actual spending or wealth profiles. The latter reflects indirect inference based on ex-post outcomes, whereas the ex-ante approach is based on matching directly with the model's behavioral predictions. In particular, heterogeneous-agent models imply that intertemporal choices under uninsured risks generate a systematic relationship between individuals' expectations of future income, perceived risk, and expected consumption growth as dictated by the Euler equation, as shown in Section 4.6. Such a relationship depends on the agent's preference parameters β and ρ. My calibration builds on recent studies that estimate Euler equations with ex-ante survey expectations to infer the intertemporal elasticity of substitution and prudence (Crump et al. 2022; Dräger and Nghiem 2021; Christelis et al. 2020). Moreover, conditioning directly on subjective income expectations allows identification of preference parameters that are otherwise unobservable—echoing Manski (2004)'s insight that expectation data enables inference without imposing indirect assumptions about beliefs.

Figure 7 compares the cross-moment fit of the baseline and the preferred subjective model when they are calibrated using indirectly inferred values of β and ρ for matching different target moments. Differences between the model-implied moments and their data counterparts are expressed in log deviations. Therefore, an octagon that spreads to coordinates of zero across all data moments indicates a perfect fit of the model to the data.

Regardless of which target moment is selected, the clearest pattern in Figure 7 is that the subjective model generally delivers a closer fit across moments than the baseline model. Both models share a common trade-off between matching wealth-to-income ratios and producing realistic H2M shares and MPCs. Parameterizations that generate more realistic average liquid wealth levels typically imply H2M shares and MPCs that are too low relative to the data. Conversely, specifications that better match H2M shares and MPCs tend to understate the extent of wealth accumulation in the economy. This trade-off is substantially weaker in the subjective model. Intuitively, this improvement

arises because separating subjective and heterogeneous beliefs from actual income realizations relaxes the tight linkage between the wealth distribution and MPCs.

FIGURE 7. Model fit across moments: baseline versus subjective model



Note: This figure shows the model fit in terms of eight population or life cycle moments based on an indirectly calibrated discount factor β and the CRRA coefficient ρ by targeting a certain data moment. All data-model differences are expressed in log deviations. For the four life-cycle moments, each axis reports the average log deviation across ages, the same per-age criterion the indirect inference minimizes. Ages at which a model-implied share approaches zero therefore dominate the corresponding axis: this is what drives the wide scale of the "Target Exp. C Life" panel, in which the baseline model's H2M share falls below 10^{-6} over much of the life cycle.

6.4.4. Heterogeneous/non-homothetic preference and borrowing constraint

One of the common additional features added to the baseline model in the existing literature to match the empirical wealth inequality is the heterogeneity in the preferences, especially in the time discount rates (Krusell and Smith 1998; Krueger et al. 2016; Carroll et al. 2017). Such a modeling assumption has been recently supported by some empirical evidence and laboratory experiments (Epper et al. 2020). Despite such indirect evidence, however, the exact degree of time preference heterogeneity in the model cannot be directly observed and estimated. Thus, the literature commonly adopts the "revealed preference" approach to indirectly calibrate the model-implied heterogeneity in preferences to match the data.

Also, the literature has increasingly relied on nonhomothetic preferences, rather than unconditional heterogeneity, to achieve the same goal, as emphasized by Dynan et al. (2004) and Straub (2019). The key mechanism is that the non-homotheticity makes saving a luxury good: as permanent income increases, one's "joy for saving" increases. This leads to a marginal propensity to consume out of permanent income that falls below one. This contrasts with the class of homothetic models considered in this paper.

Rather than microfounding such preferences, I adopt a reduced-form approach by recalibrating ex-ante preference parameter heterogeneity to replicate similar qualitative patterns. Specifically, I study three extensions of the subjective model in which agents with steeper life-cycle income growth are also endowed with preference parameters that imply systematically higher saving rates. These agents receive a higher discount factor β, a higher coefficient of relative risk aversion ρ, and a stronger bequest motive near the end of life. All other income processes and calibrations remain identical to

those in the baseline subjective model. From low to high growth types, their respective values are $\rho = \{1.1, 2.0, 3.0\}$; $\beta = \{0.94, 0.96, 0.99\}$; $\phi = \{0.02, 2.0, 5.0\}$. The three models' results are reported in Table 5, under the names of *CRRA*, *Patience*, and *Bequests*.

The income-dependent preference heterogeneity further widens the wealth distribution relative to the subjective model, particularly by increasing the savings of the wealthy in the model. All three models increase the wealth Gini by 3, 2, and 1 points, respectively. The H2M share also increases by 3, 1, and 2 percentage points, respectively, and the average MPC changes from 0.25 to 0.27, 0.25, and 0.25, respectively. Meanwhile, the three models raise the average wealth-to-income ratio from 0.68 to 0.94, 0.86, and 0.73, respectively. Consistent with the literature, the introduction of non-homothetic preferences has the merit of generating wealthier households in the model while maintaining a good fit at the lower end of the distribution. Since this paper emphasizes the role of ex-ante heterogeneity in permanent income profiles across agents, incorporating non-homothetic preferences serves as a natural complementary feature to improve the model fit.

Lastly, it is also natural to introduce another realistic feature that turns out to be complementary to my subjective model. In particular, I relax the zero-borrowing constraint in Equation 16 so that the agents in my model can actually borrow up to their "natural borrowing constraint". What is interesting in this alternative environment is the fact that the natural borrowing constraint is actually dependent on the perceived unemployment risks. Unsurprisingly, as reported in the "Borrowing" model in Table 5, the wealth distribution shifts leftward, which increases the wealth Gini by 5 percentage points, the share of H2M by a substantial 16 percentage points, and yields a slightly lower average MPC of 0.22. Intuitively, borrowing capacity breaks the usual link between H2M status and MPCs: many agents now optimally hold near-zero or negative liquid wealth without being at their effective constraint, and the option to borrow flattens the consumption function at low wealth, so measured H2M rises sharply while the average MPC does not.

Taken together, these extensions show that adjusting subjective and heterogeneous income risks is not only consistent with, but in fact reinforces, the additional model ingredients typically employed in the literature to improve the empirical fit of the standard heterogeneous-agent life-cycle model. That said, even though my subjective model substantially better replicates the empirical patterns, a considerable discrepancy remains, one that the extra features are still required to address. At the same time, it is arguably natural that, once the observed dispersion in income expectations and risk perceptions is explicitly taken into account, the degree of preference heterogeneity needed to fully "bridge the gap" is correspondingly reduced.

6.5. Final discussion

Before concluding, I briefly discuss several other issues related to the robustness of the key insight of this paper and some future research directions.

The focus on wage and unemployment risks. Although this paper focuses on the perception patterns related to two specific sources of labor income risks—wage changes and unemployment transitions—the central conclusion likely extends to total earnings and household dynamics more broadly. Since it is a common practice in the incomplete market macroeconomic models to assume inelastic labor supply, this paper implicitly takes a broad interpretation of the wage risks as on-the-job earning risks. Subjective

earning risks, when measured broadly, exhibit patterns in line with those reported in this paper. For example, Caplin et al. (2023) demonstrate that the holistic subjective earning distribution, constructed from separately elicited job-transition probabilities and conditional earnings distributions, displays the same qualitative pattern as in this study: perceived risks are highly heterogeneous between individuals and are lower than the dispersion of actual income changes. That said, the survey measures used here have their limitations. Most notably, they do not capture perceived risks regarding working hours, a potentially important margin of earnings variation; I leave this for future work.

A more general treatment of the heterogeneity in perceived income processes. This paper's treatment of heterogeneity in subjective income processes is limited to the heterogeneity in a small set of parameters for a commonly adopted income process. In the Appendix Figure A.3, I adopt a machine learning approach to cluster the subjective distribution patterns into 10 groups. The clustered distributions are widely heterogeneous. But across all types, the dispersion of the subjective distribution remains in a lower range of values than the calibrated risks, consistent with my main finding. It would certainly be fruitful in future research to incorporate richer heterogeneity in perception patterns using non-parametric methods into the heterogeneous-agent model considered in the paper.

Non-Gaussian income processes. A large body of work—much of which uses administrative income records (Guvenen et al. 2021)—shows that labor earnings dynamics are not well captured by the standard Gaussian process. It is worth clarifying that the results in this paper on the discrepancy between perceived and calibrated risks are unlikely to be sensitive to considering non-Gaussian income processes for the following two reasons.

Most importantly, the literature (Guvenen et al. 2021) shows that deviations from Gaussian dynamics, i.e., skewness, largely arise from discrete job transitions that are embedded in observed earnings changes. In this respect, this paper's analysis explicitly incorporates such risks through separately calibrated job transition probabilities. Second, although a richer specification of the stochastic process is proven to better characterize the earnings dynamics, it does not rule out the possibility that income risks are perceived differently by the agents for the reasons discussed in this paper. With that said, it would be a fruitful direction of research to combine the subjective earning distributions with high-quality income records to uncover richer earning/income dynamics. (See Arellano et al. (2024) for a recent example, and Arellano et al. (2026), who use the subjective expectations in the same SCE to estimate earnings, job, and employment dynamics.)

General-equilibrium. The core insights of this paper are already well captured by the partial-equilibrium life-cycle framework. It is, however, natural to ask whether the quantitative results obtained so far continue to hold in a setting where the real interest rate is determined endogenously through market-clearing in asset markets. In Appendix B.4, I therefore analyze a one-asset general-equilibrium overlapping-generations model in the spirit of Huggett (1996); Carroll et al. (2017); Krueger et al. (2016), in which household savings serve as productive capital. The main conclusion is that, in general equilibrium, all central results of the paper continue to hold despite attenuation of varying degrees. I leave it for future research to explore richer implications of perceived income risks once additional important features such as portfolio choices between risky/illiquid assets are introduced, as in Kaplan et al. (2014).

7. Conclusion

Incomplete-market macroeconomic models featuring uninsured idiosyncratic income risks rarely incorporate the heterogeneity in income risks as perceived by agents. Using density forecasts of wage growth and job-transition probabilities from the New York Fed's *Survey of Consumer Expectations*, I document two facts and explore their model implications. First, perceived risks are more heterogeneous than assumed in common calibrations, providing an additional observable input for matching the empirical wealth inequality. Second, perceived risks are lower than conventional calibrations; the gap reflects anticipated heterogeneity and an under-perception of risk relative to the income process taken as objective, which I refer to as overconfidence. This helps explain why these models usually predict higher buffer-stock savings than found in the data.

This paper demonstrates the rich potential of using survey data in calibrating heterogeneous-agent models where realistic heterogeneity in expectations/perceptions yields better model predictions. With increasingly available survey data on decision-relevant expectations, economists are no longer confined to calibrating important model parameters, such as income risks, indirectly from panel data under the stringent assumption of rational expectations. The use of survey-implied heterogeneity establishes a direct link between expectations and behaviors and helps economists do a better job of matching empirical patterns within the macroeconomy. Even without such survey data, this paper invites separating the risks agents perceive from the risks that actually shock the population—a likely candidate for bringing heterogeneous-agent models closer to the data.

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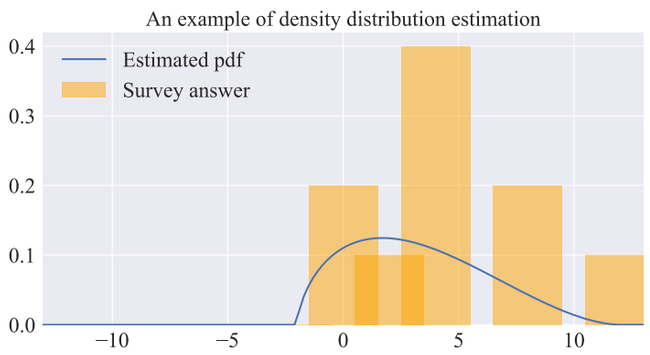
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Appendix A. Additional empirical results

A.1. Density estimation of the survey answers

With the histogram of the answers for each individual in hand, I follow Engelberg et al. (2009) to fit each of these answers with a parametric distribution accordingly for the three following cases (see Figure A.1 for an example). In the first case, when three or more intervals are filled with positive probabilities, these are fitted with generalized beta distributions. In particular, if there is no open-ended bin on the left or the right, then a two-parameter beta distribution is sufficient. If there is an open-ended bin with a positive probability on either the left or the right, since the lower or upper bound of the support needs to be determined, a four-parameter beta distribution is estimated. In the second case, in which there are exactly two adjacent intervals with positive probabilities, this is fitted with an isosceles triangular distribution. In the third case, if there is only one interval with positive probability, that is, a probability equal to one, this is fitted with a uniform distribution.

FIGURE A.1. An illustration of the density estimation of the survey answers



Note: This is one example of a bin-based forecast of the wage growth in the *Survey of Consumer Expectations* (SCE) and how it is fit by a parametric distribution. The horizontal axis shows the values of the expected wage growth and the vertical axis shows the probabilities assigned by the respondents.

For all of the moment estimates, there are inevitably extreme values. This could be due to the idiosyncratic answers provided by the original respondents, or some non-convergence of the numerical estimation program. Therefore, for each moment of the analysis, I exclude the top and bottom 1% observations, leading to a sample size of around 53,180.

A.2. Other facts about wage expectations and perceived risks

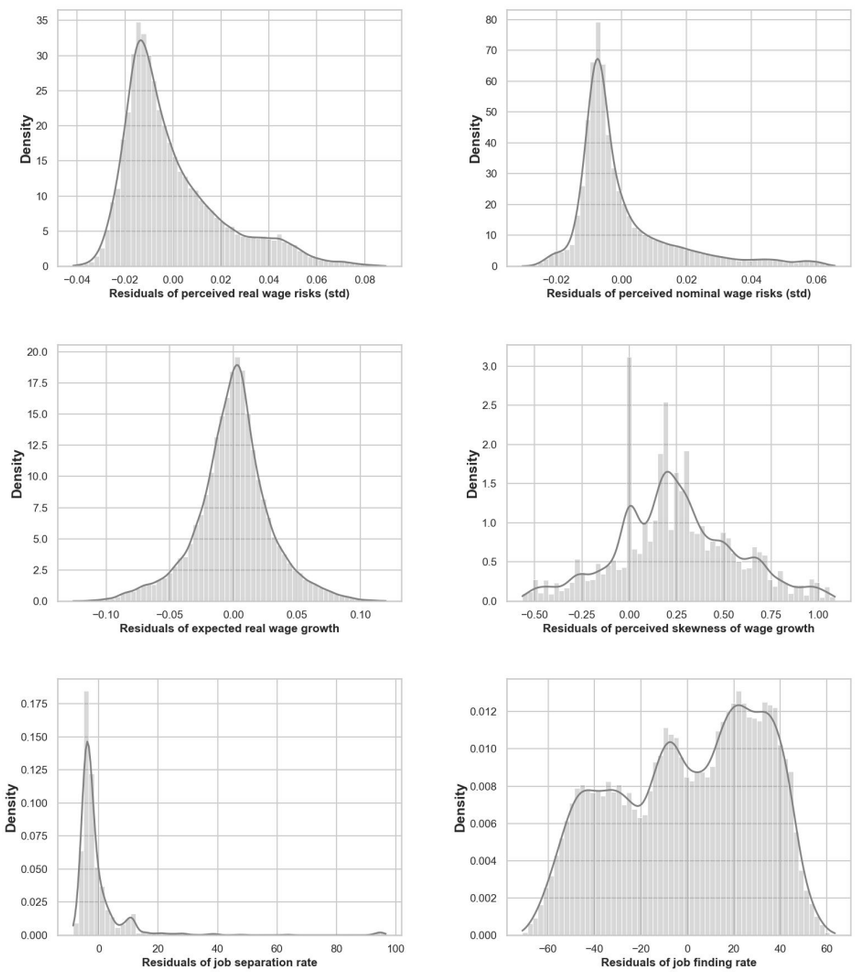
A.2.1. Heterogeneity of perceptions in different moments

Figure A.2 shows the within-group heterogeneity of real PRs, nominal PRs, expected real wage growth rates, perceived skewness, perceived job separation, and perceived job finding probabilities, controlling for observable demographic variables in the SCE.

A.2.2. Perceived risks and observable characteristics

Table A.1 regresses the perceived risk of real wage growth on the observable characteristics available in the SCE, using the same age and education groupings as Table 1. It

FIGURE A.2. Dispersion in expected wage growth and perceived skewness



Note: The distributions of the residuals of the real and nominal perceived risks (PRs) (in standard deviation terms), the expected real wage growth rate, perceived skewness of the 1-year-ahead wage growth, perceived job separation, and job finding probabilities in the SCE that are unexplained by observable demographic variables.

quantifies the group differences that Table 1 and Figure 2 display as averages. Standard errors are clustered by household.

The covariates are jointly significant at any conventional level in every specification, but they explain only a small share of that dispersion: age, gender, and education together account for 1.9% of the variance of perceived risks, and every observable in the survey for no more than 5.2%.

A.2.3. Machine learning clustering of survey answers

Density forecasts of wage growth in the SCE used in this paper's analysis are widely heterogeneous. In addition to indirectly working with the distribution across a set of estimated moments, I also employ a k-means machine learning algorithm to cluster the density forecasts into heterogeneous types. Figure A.3 plots the average histograms of the 10 types of subjective distribution. The number of clusters, 10, is chosen via k-fold cross-validation. For each type, I report the share of observations belonging to that type, the mean, and the standard deviation based on the same density estimation procedure as in Section A.1. The largest standard deviation across types is 0.08, and the type displaying it accounts for only 5.3% of the SCE observations. Nine out of the

TABLE A.1. Perceived wage risks and observable characteristics

	(1)	(2)	(3)	(4)	(5)
Age 25-29	-0.597***	-0.498***	-0.505***	-0.391***	-0.414***
	(0.154)	(0.153)	(0.153)	(0.151)	(0.151)
Age 30-34	-0.817***	-0.721***	-0.731***	-0.587***	-0.624***
	(0.154)	(0.153)	(0.153)	(0.151)	(0.151)
Age 35-39	-0.855***	-0.776***	-0.782***	-0.618***	-0.724***
	(0.154)	(0.153)	(0.153)	(0.151)	(0.151)
Age 40-44	-0.946***	-0.890***	-0.895***	-0.728***	-0.778***
	(0.154)	(0.153)	(0.154)	(0.152)	(0.152)
Age 45-49	-0.943***	-0.896***	-0.900***	-0.739***	-0.810***
	(0.154)	(0.153)	(0.153)	(0.151)	(0.151)
Age 50-54	-1.043***	-1.038***	-1.050***	-0.895***	-0.986***
	(0.153)	(0.152)	(0.153)	(0.150)	(0.150)
Age 55-60	-1.049***	-1.041***	-1.051***	-0.922***	-1.051***
	(0.153)	(0.153)	(0.153)	(0.150)	(0.150)
Male		0.155***	0.155***	0.079**	0.147***
		(0.040)	(0.040)	(0.040)	(0.040)
HS dropout		0.594	0.605	0.450	0.837*
		(0.419)	(0.417)	(0.402)	(0.442)
College or above		-0.456***	-0.443***	-0.307***	-0.336***
		(0.044)	(0.044)	(0.044)	(0.045)
Low numeracy			0.182***	0.164***	0.166***
			(0.045)	(0.044)	(0.046)
Low household income				0.679***	0.631***
				(0.037)	(0.038)
Part-time				0.168***	0.069
				(0.054)	(0.060)
Perceived job-separation prob.					0.005***
					(0.001)
Perceived agg. unemployment prob.					0.006***
					(0.001)
N	52823	52823	52823	52823	47654
Households	9955	9955	9955	9955	9011
R2	0.008	0.019	0.021	0.043	0.052
F (all covariates)	13.0	22.9	22.3	44.0	43.4
SD explained (%)	8.7	13.9	14.4	20.7	22.9

Standard errors, in parentheses, are clustered by household. * p<0.1, ** p<0.05, *** p<0.01.
The dependent variable is the perceived risk of one-year-ahead real wage growth, in standard-deviation units and percentage points – the same object reported in Table 1 and Figure 2. Its mean is 3.02 and its standard deviation 2.19 in the estimation sample.
Age groups use the same five-year cuts as Table 1, with ages 20–24 omitted; age 60 is folded into the top group rather than dropped. Education groups also follow Table 1, with high-school graduates omitted.
Columns (1)–(4) are estimated on a common sample. Column (5) adds the subjective job-separation and aggregate-unemployment expectations, which the SCE elicits only from employees, and so rests on a smaller sample that excludes the self-employed.
"SD explained" is the standard deviation of the fitted values as a percentage of the standard deviation of the dependent variable. Together with R^2 it shows that observable characteristics account for only a small share of the cross-sectional dispersion in perceived risks.

ten types have an implied standard deviation that is less than 5%. This assures that the finding that perceived risks are lower than calibrated risks, as reported in the main text, is not simply driven by an unusual subsample's behaviors.

A.3. Wage risk estimation using SIPP data

A.3.1. Sample selection

To estimate the wage risks or risks to earnings, conditional on working the same hours and staying in the same job, I restrict the analysis to the *Survey of Income and Program Participation* (SIPP) sample based on such a definition for the worker's primary job (JB1). The specific filtering criteria are listed below, and these are approximately identical to those in Low et al. (2010) for computing the wage rate for the same job, using 1993 panel data from the SIPP.

- Time: March 2013–December 2021
- Age: 20-60 years old
- Work arrangement: employed by someone else (excluding self-employment and other work arrangements): EJB1_JBORSE ==1.
- Employer: never changing employer over the observed spell, and observed for a tenure of at least two years: a constant EJB1_JOBID for 24 or more months.
- Wage: total monthly earnings from the primary job divided by the average number of hours worked in the same job, wage = TJB1_MSUM/TJB1_MWKHRS.
- Outliers: drop observations with wage rate lower than 0.1 or greater than 10 times the individual's average wage, then drop the top and bottom percentile of the wage distribution within each month.
- No days off without pay: EJB1_AWOP1 = 2.
- Continued job spell since December of the last year: RJB1_CFLG=1.
- Drop imputed values: EINTTYPE==1 or 2.
- Drop government/agriculture jobs: drop if TJB1_IND>=9400.

Based on the selected sample, Table A.2 reports the size and approximated group-specific wage volatility as defined in Equation 4.

A.3.2. Seam effect

One special feature of the SIPP is that it collected monthly information by surveying each respondent every four months before the 2013 wave and once a year afterward (since the 2014 wave). This leads to the well-documented issue of the seam effect (Ryscavage 1993; Rips et al. 2003; Nekarda 2008; Callegaro 2008), whereby reported changes in survey answers are relatively small for adjacent months within a survey wave but changes are much more abrupt between months across surveys. Such a difference could be either due to the under-reporting of changes within a reference period (for reasons such as the recall bias) or the over-reporting of changes across the reference periods.

FIGURE A.3. K-means clustering of the density forecast of wage growth in the SCE

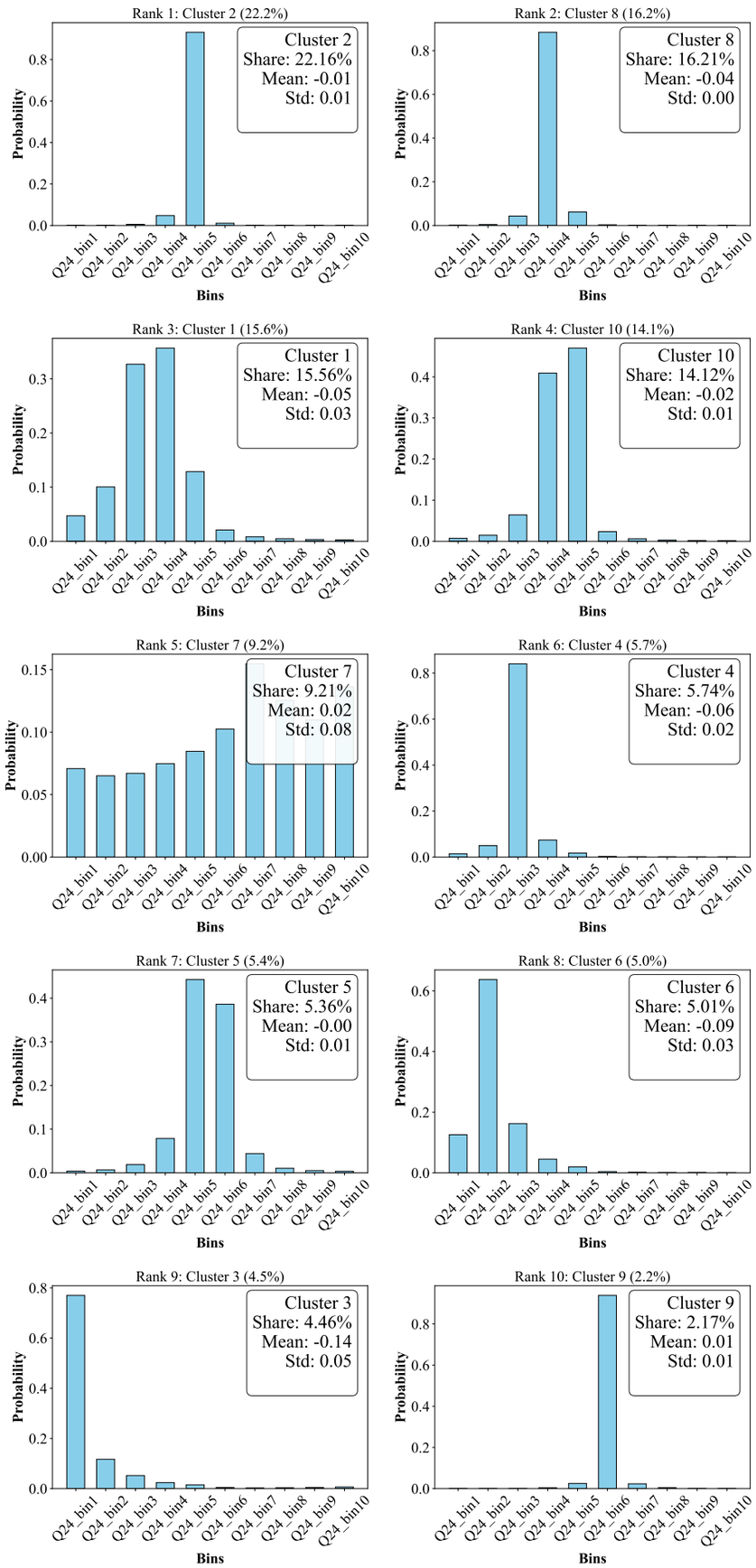


TABLE A.2. Summary statistics of the SIPP sample

	Obs	Volatility
Year		
2013 (14%)	9,278	N/A
2014 (16%)	12,011	0.41
2015 (12%)	8,853	0.37
2016 (8%)	5,699	0.34
2017 (9%)	6,305	N/A
2018 (11%)	7,877	0.45
2019 (11%)	8,047	0.37
2020 (10%)	7,131	0.35
2021 (4%)	2,974	0.42
Education		
HS dropout (21%)	14,900	0.39
HS graduate (45%)	31,345	0.39
College/above (33%)	21,930	0.39
Gender		
male (56%)	38,181	0.38
female (43%)	29,994	0.4
Total (100%)	68,175	0.39

This effect is clearly seen from the time series plot of monthly wage volatility in Figure A.4, where there is always a spike in the size of the volatility between December and January, in the sample period.[20]

Due to this seam issue, for the monthly risk estimations, I exclude observations for December and January, leading to the non-identification of the risks of each January. By doing so, I effectively assume that the within-wave respondents did not under-report the true changes to their wages, while the cross-wave answers over-reported these changes. But the opposite assumption might be true, in that respondents under-reported the changes within the reference year when they retroactively answered the survey questions, and the changes across the reference periods were correctly reported.

One way to incorporate the cross-wave changes instead of dropping them by brute force is to estimate the risks at lower frequencies, that is, quarterly and yearly, and to construct the quarterly/yearly periods such that these cover the cross-wave cutoff month in December.

A.4. Wage risk estimation under alternative assumptions

A.4.1. Baseline estimation

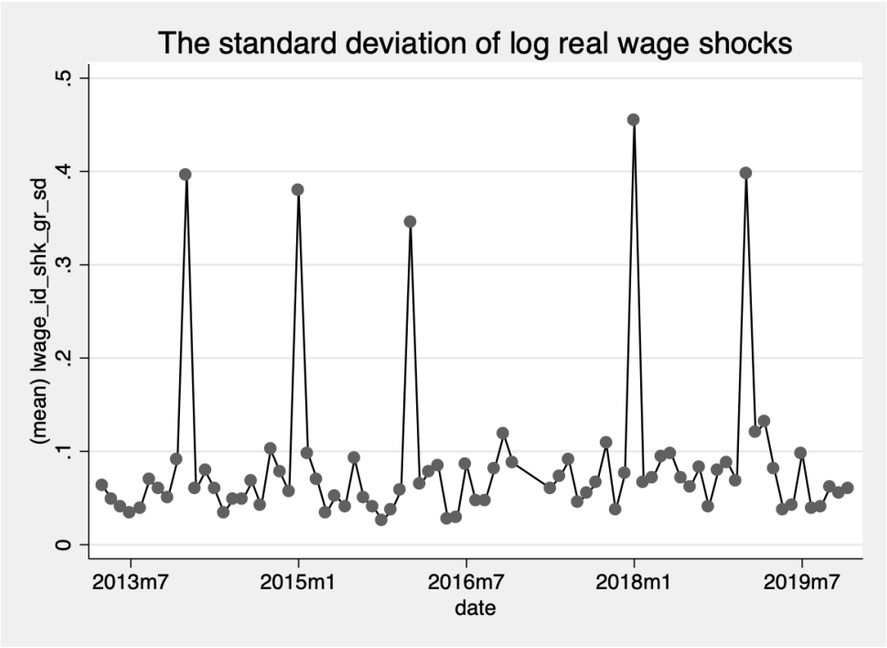
Permanent and transitory risks are identified via the following moment restrictions.

$$
\begin{aligned}
&var(\Delta e_{i,t}) = var(\psi_t + \theta_t - \theta_{t-1}) = \sigma^2_{\psi,t} + \sigma^2_{\theta,t} + \sigma^2_{\theta,t-1} \\
&cov(\Delta e_{i,t}, \Delta e_{i,t+1}) = cov(\psi_t + \theta_t - \theta_{t-1}, \psi_{t+1} + \theta_{i,t+1} - \theta_{i,t}) = -\sigma^2_{\theta,t} \\
&cov(\Delta e_{i,t-1}, \Delta e_{i,t}) = cov(\psi_{t-1} + \theta_{t-1} - \theta_{t-2}, \psi_t + \theta_{i,t} - \theta_{i,t-1}) = -\sigma^2_{\theta,t-1}
\end{aligned}
\tag{A1}
$$

With four consecutive observations of the wages of individual i from $t-2$ to $t+1$, hence, three observations of the first residual difference Δe, the above three equations

[20] Note that the only exception is for January 2017, for which no monthly growth rate is available due to the reshuffling of the SIPP sample.

FIGURE A.4. Estimated monthly wage volatility



Note: The monthly wage volatility as defined in Equation 4 for the entire selected sample, estimated from the SIPP.

can exactly identify the permanent risk specific to time t, $\sigma_{\psi,t}$, and the time-specific transitory risks $\sigma_{\theta,t}$ and $\sigma_{\theta,t-1}$.

Three consecutive observations of wage data are sufficient under a looser restriction that the transitory risks stay constant over each 3-period horizon, between $t-1$ and $t+1$, call it $\bar{\sigma}_{\theta,t}$. In particular, the identification is as follows.

$$\begin{aligned} var(\Delta e_{i,t}) &= var(\psi_t + \theta_t - \theta_{t-1}) = \sigma^2_{\psi,t} + 2\bar{\sigma}^2_{\theta,t} \\ cov(\Delta e_{i,t}, \Delta e_{i,t+1}) &= -\bar{\sigma}^2_{\theta,t} \end{aligned} \tag{A2}$$

Figure A.5 plots the identified time-varying component-specific risks under a wage process set at monthly frequencies. These are used to compute the calibrated wage risks in Table 1 and Figure 2.

A.4.2. From monthly risks to the annual comparison object

The perceived risk elicited in the SCE is a *point-in-time* object: as of month t, the respondent reports the distribution of their wage twelve months ahead relative to its current level, conditional on staying in the same job. The calibrated counterpart reported in Table 1 is constructed to match that object exactly. This subsection derives it from the monthly estimates obtained above.

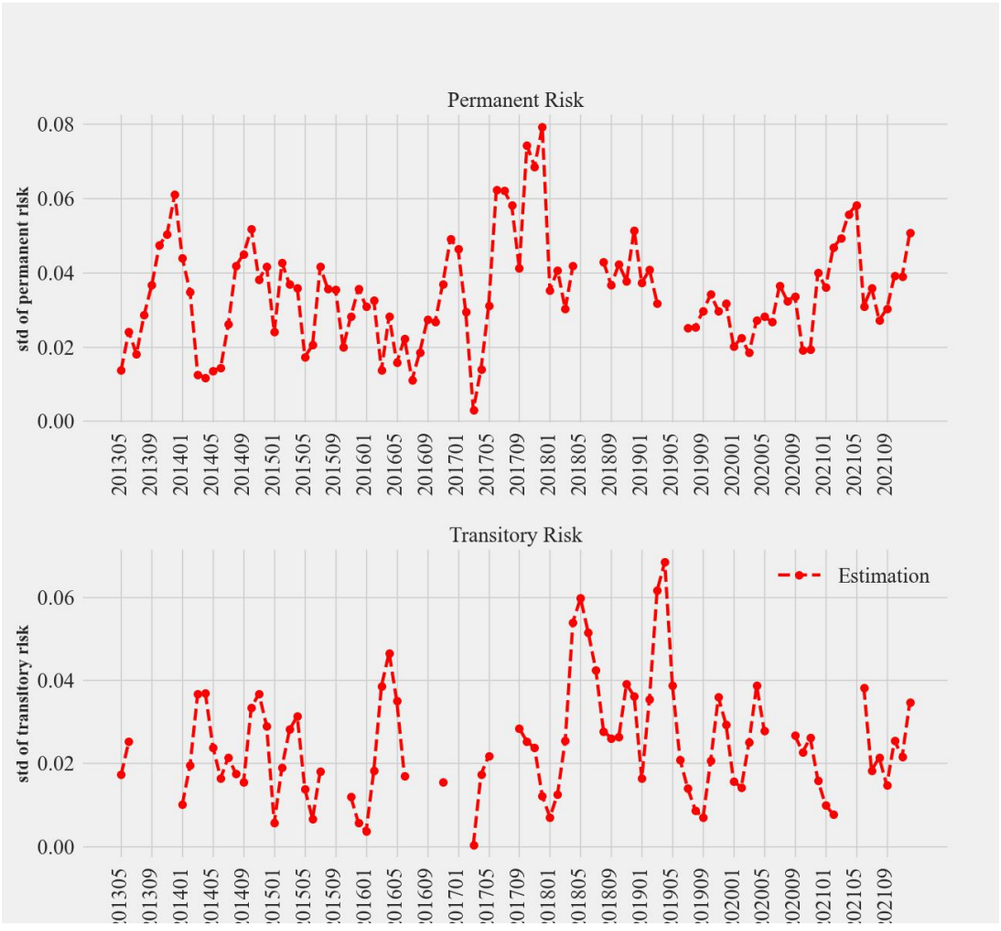
Let the monthly wage process be

$$y_t = p_t + \theta_t, \tag{A3}$$

$$p_t = p_{t-1} + \psi_t, \tag{A4}$$

where ψ_t is a permanent (random-walk) innovation with variance $Var(\psi_t) = \sigma^2_\psi$ and θ_t is an iid transitory shock with variance $Var(\theta_t) = \sigma^2_\theta$.

FIGURE A.5. Monthly permanent and transitory wage risks



Note: The 3-month moving average of the estimated monthly permanent and transitory risks (in standard deviation units), using SIPP panel data on wages from March 2013 to December 2021.

The twelve-month log wage change is then

$$y_{t+12} - y_t = \sum_{j=1}^{12} \psi_{t+j} \; + \; \theta_{t+12} - \theta_t. \tag{A5}$$

Two features determine the mapping to annual frequency. First, the twelve permanent innovations realized between t and $t+12$ are mutually independent, so their variances simply add. Second, the transitory component of the *current* wage, θ_t, is already known to the respondent at the time of the forecast—it is embedded in the wage level they are asked to forecast from—and so contributes nothing to the conditional variance. Only the future draw θ_{t+12} is uncertain. Hence

$$Var_t\left(y_{t+12} - y_t\right) = \sum_{j=1}^{12} \sigma^2_{\psi,t+j} \; + \; \sigma^2_{\theta,t+12}. \tag{A6}$$

Implication.. The annual permanent variance is the *sum* of the twelve monthly permanent variances over the forecast window—twelve times the monthly average if risks are constant within the year—while the annual transitory variance is that of a *single* monthly transitory shock. In implementing this I use the average of the twelve monthly transitory variances over the window rather than the twelfth one alone: the two are equivalent under within-year constancy, and the average is less sensitive to month-specific estimation error. This is the mapping used to produce the annualized risks in Table 1.

It is worth distinguishing this conditional variance from the *unconditional* variance of a realized twelve-month change, $Var(y_{t+12} - y_t) = 12\sigma^2_\psi + 2\sigma^2_\theta$, in which both endpoints' transitory draws are treated as unknown. The factor of two on the transitory term is appropriate for a realized change but not for a forecast made from an observed current wage, which is the object the survey elicits. Since the estimated transitory risks are an order of magnitude smaller than the permanent ones at monthly frequency, the distinction is quantitatively minor here—it raises the total annual calibrated risk by about 0.3 percentage points in standard-deviation units—but it matters for the interpretation of the comparison.

A.4.3. Estimated wage risks at a lower frequency

Most of the income risk estimation in the literature is done at lower frequencies, such as yearly and quarterly.

Year-over-year wage growth can be constructed for 2015, 2016, and 2019–2021. Holding the transitory variance constant within each block of consecutive years, this identifies the year-specific permanent risks for those five years and the average transitory risks for 2015–2016 and 2019–2021. The two gaps reflect the structure of the SIPP sample: no individual is observed across the 2016/2017 changeover, when the entire sample was replaced, and the first year of a sample yields no year-over-year growth.

The estimated sample averages are reported in Table A.3. For the years with identified risks, the estimated risks at annual frequencies seem to be much larger than those commonly seen in the literature, as summarized in Table A.4. In particular, the size of the permanent shock is estimated to be 32%, in contrast to the standard estimates of 10-15%. The transitory risks are estimated to be around 23%, exceeding the standard

TABLE A.3. Estimated wage risks at lower frequencies

Yearly Permanent	0.324408
Yearly Transitory	0.233421
Quarterly Permanent	0.313773
Quarterly Transitory	0.120022

Note: expressed in standard deviation units. For yearly estimates, the year-over-year growth of monthly wage rates is used.

estimates of 10% to 20%.

A similar pattern can be seen from the quarterly estimates using quarterly growth of average wage rates (see also Table A.4).

A.5. Perceived versus calibrated risks: robustness

A.5.1. A comparison of simulated wage growth from density survey and realized wage growth

A complementary approach to evaluate wage growth expectations and perceived risk is to simulate the distribution of wage growth implied by the full set of individual density forecasts. For each SCE respondent, I draw 1,000 realizations directly from their reported wage-growth histogram and pool these draws to form the implied distribution. This approach is agnostic to any parametric mapping from histogram responses to underlying beliefs. The resulting dispersion reflects both heterogeneity in expected wage growth across individuals and the average perceived risk. I also construct a counterfactual distribution that isolates heterogeneity in expectations by abstracting from perceived risk. Comparing these simulated distributions with the realized wage growth distribution in the SIPP reveals a clear pattern: the survey-implied dispersion is far smaller than the actual dispersion in the SIPP, and the version based solely on expectational heterogeneity is even slightly narrower, underscoring the modest perceived risks in the survey data.

A.6. Income risks in the existing literature

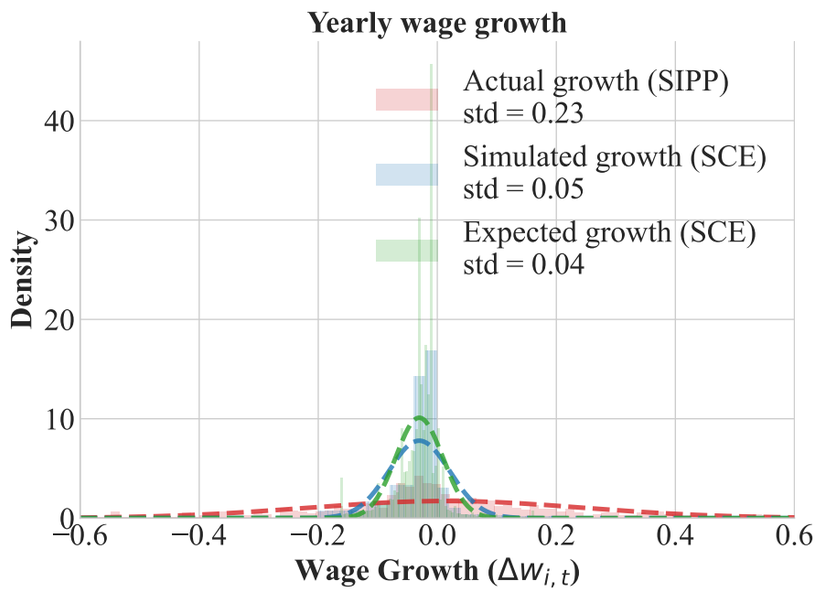
Table A.4 summarizes the most common estimates of income risks seen in the literature. For comparison, I convert all of the risks into annual frequencies (because some of the estimates are for different frequencies). Whenever group-specific risks are assumed (depending on education and age), I summarize them as a range. Also, for models that assume a persistent instead of a permanent component, I treat the assumed size of the persistent risks as a lower bound for the permanent risks.[21] For models with income risks dependent on aggregate business cycles, à la Krusell and Smith (1998), I compute the steady-state size of the idiosyncratic risks by using the transition probabilities of the aggregate economy employed in the paper.

A.7. Comparison of perceived and calibrated unemployment risks

Figure A.7 compares the time-series pattern of the perceived unemployment risks and calibrated risks over the sample period. As in the main body, I cast both probabilities

[21] One can think of the permanent income shock as a limiting case of the AR(1) shock, with the persistence parameter infinitely close to 1. The effective income risks increase with the persistence of the shock.

FIGURE A.6. Simulation of wage growth rate: survey versus panel



This figure compares three wage-growth distributions: (i) the simulated distribution based on 1,000 draws from each respondent's SCE histogram forecast, (ii) the actual wage-growth distribution from the SIPP, and (iii) the distribution of expected wage growth reported in the SCE. Each distribution is also shown with its corresponding normal fit.

into a continuous-time rate for a Poisson point process, assuming constant hazard.

Two patterns emerge. First, although the two series are constructed independently of one another, on average, the perceptions track the aggregate realizations relatively well. The most notable deviation between the beliefs and the realizations occurred during March 2020, which saw an unprecedented increase in one-month job separations and a dramatic decrease in job finding. Second, however, as shown by the wide 25/75 interquartile range around the mean expectations, individual respondents vastly disagree on their individual separation and finding probabilities. Because the question in the survey concerns individual-specific transitions, it is reasonable to assume that this reflects either the unobserved heterogeneity or the information available on each individual's status, which economists cannot directly observe.

Appendix B. Additional model results

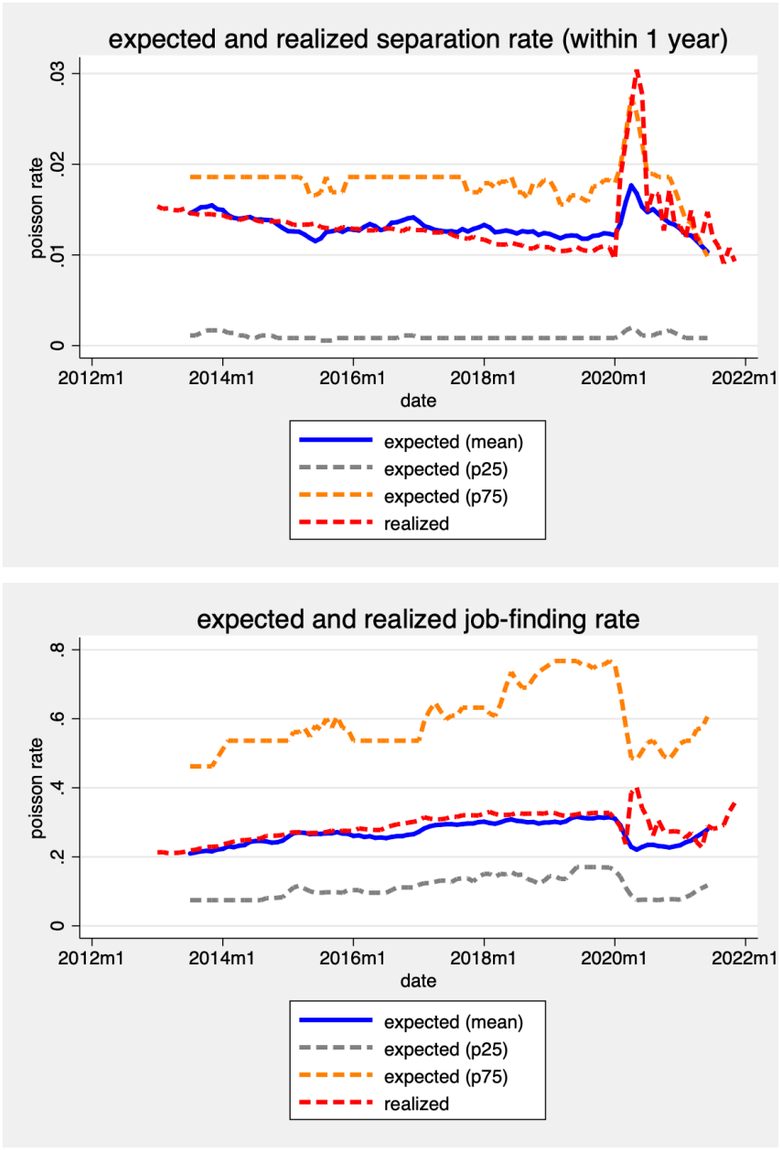
B.1. Homogeneous and heterogeneous life-cycle wage profiles

Figure A.8A plots the deterministic wage profile used to calibrate the baseline model, which is estimated based on the SCF total wage income. Figure A.8B plots the heterogeneous wage profiles calibrated to the subjective model. Figure A.9 plots the heterogeneous wage profiles used in other model experiments, which are calibrated based on the heterogeneous wage growth rates reported in the SCE.

B.2. Other model moments

Figure A.10 plots the other model-implied moments relative to their data counterparts, whenever available. They include the distribution of MPCs across agents, the average life-cycle profile of the expected consumption growth factor, and the distribution of the

FIGURE A.7. Expected and realized job-separation and finding rates



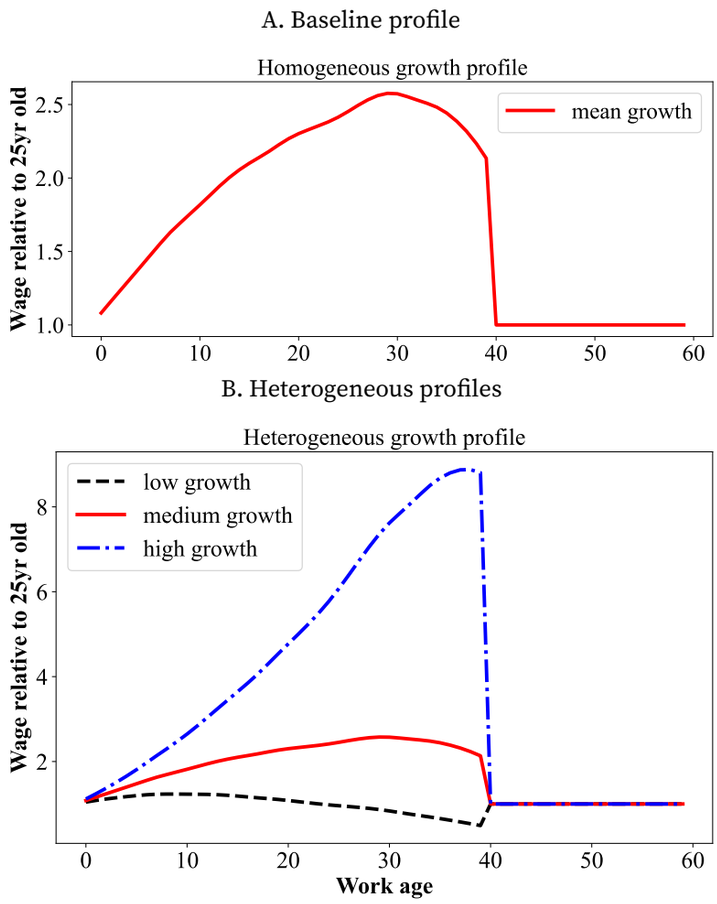
Note: Realized job-separation and finding rates are computed from the CPS following the method of Fujita and Ramey (2009), which is directly available at the Federal Reserve Bank of San Francisco. Both the realizations and the perceived probabilities are expressed as Poisson point rates in continuous time, with one month as the unit of time. The 3-month moving average of each series is plotted.

expected consumption growth factor.

B.3. Model results: robustness

Table A.5 reports the comparison of baseline and subjective model moments under alternative parameter values. In the first case, I use a pair of smaller values of the calibrated permanent and transitory risks, $\sigma_\psi = 0.10$ and $\sigma_\theta = 0.10$, than those used in the baseline. In the second case, I use two alternative unemployment insurance replacement ratios, $\zeta = 0.15$ and $\zeta = 0.6$, one below and one above the baseline value of 0.5. Moving from $\zeta = 0.15$ to 0.5 to 0.6, the subjective model raises the wealth Gini relative to the baseline by 16, 11, and 11 percentage points, and the H2M share by 22, 22, and 20 percentage points. A more generous replacement ratio therefore narrows the gain in wealth inequality, since it raises the baseline's own inequality, but leaves the gain in the hand-to-mouth share essentially intact. In the third case, I alter the relative ratio of subjective perceptions of permanent and transitory risks: $\kappa = 0.1$ and $\kappa = 10$, respectively. In the fourth case, I vary how the heterogeneous life-cycle growth profiles $\mathbb{G}$ are discretized: five

FIGURE A.8. Estimated deterministic wage profile over the life cycle



Note: The top and bottom panels, respectively, plot the homogeneous and heterogeneous life cycle profiles of permanent wage growth. The homogeneous profile is based on a regression of fourth-order age polynomials of the real wage income in 2016 SCF, controlling for education, occupation, gender, etc. The post-retirement profile is assumed to stay flat after a one-time drop. The heterogeneous profiles are calibrated to be consistent with the estimated heterogeneity $\sigma_{\xi,\psi} = 0.028$.

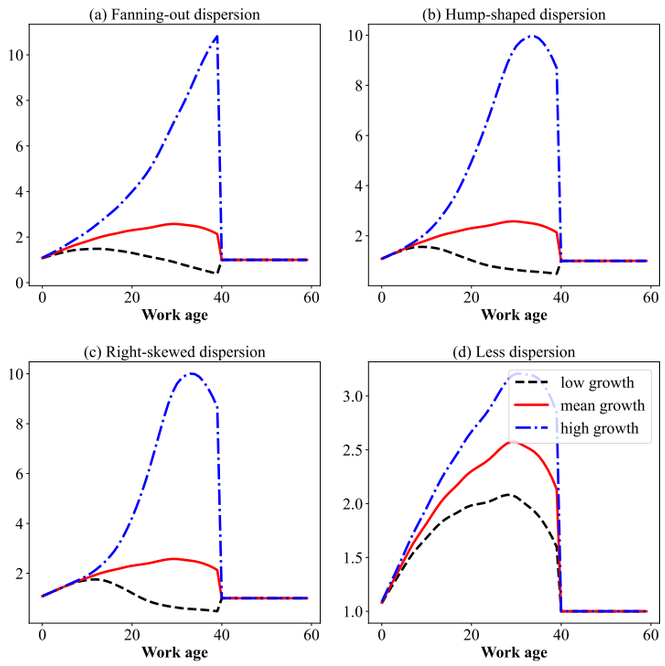
equiprobable profiles instead of three, and Gauss-Hermite quadrature—which chooses both nodes and probability weights to match higher moments—with three and five nodes. The wealth Gini ranges from 0.82 to 0.84 across these four discretizations, the H2M share from 0.29 to 0.31, and the average MPC from 0.24 to 0.25. The two schemes differ most with three nodes and nearly coincide with five, as both approximate the same underlying distribution. In each case, the improvement of the subjective model's empirical fit relative to the baseline model remains significant.

B.4. General equilibrium implications

To complement the model results in the main text, which focus on partial-equilibrium effects, this section provides additional results within a general-equilibrium framework, in the spirit of Huggett (1996); Krueger et al. (2016); Carroll et al. (2017). In this environment, production is endogenous and households' savings serve as the capital input in production. Both the real interest rate and the wage are determined within the model. Beyond the life-cycle consumption–saving problem described in the main text, the remaining elements of the model are described below. I assume there is no aggregate uncertainty in the economy, unlike in Krusell and Smith (1998).

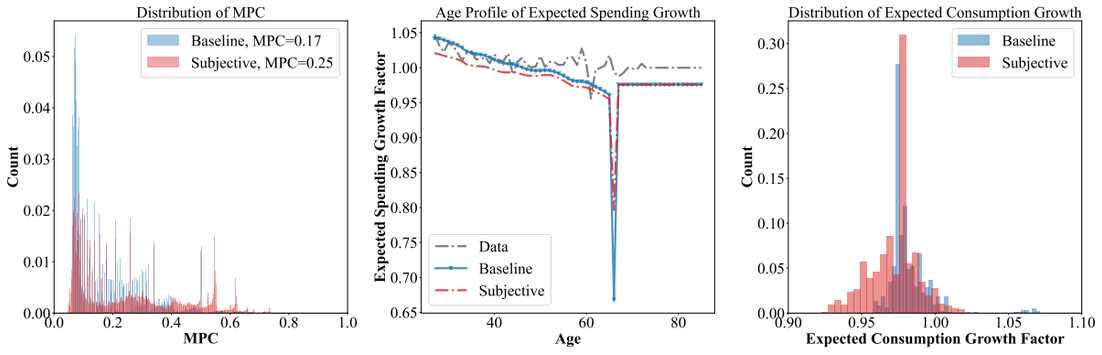
Technology. The economy has a standard constant-returns-to-scale technology that turns capital and the supplied efficient units of labor into aggregate output.

FIGURE A.9. Heterogeneous wage profiles over the life cycle



Note: These figures plot the three equally probable heterogeneous deterministic wage profiles that are used to calibrate different model specifications.

FIGURE A.10. Model versus data: other moments



$$Y = ZK^{\alpha}N^{1-\alpha} \tag{A7}$$

Capital depreciates at a rate of δ_K each period. The factors of the input markets are fully competitive.

In terms of calibration, the annual depreciation rate is set to be $\delta_K = 2.5\%$. The capital share takes a standard value of $\alpha = 0.36$ for the U.S. economy. Without aggregate shocks, Z is simply a normalizer. Therefore, I set its value such that the aggregate wage rate W is equal to one under a capital/output ratio of $K/Y = 3$ at the steady-state level of employment in the model.

Government. The government runs a balanced budget in each period. Therefore, outlays from unemployment insurance are financed by an income tax that is levied on both labor income and unemployment benefits. Given a replacement ratio ζ and the proportion of the employed population $1 - \Pi^{U}$, the corresponding tax rate λ can be easily pinned down based on the equation below.

$$\lambda \left[1 - \Pi^{\mho} + \zeta \Pi^{\mho}\right] = \zeta \Pi^{\mho} \tag{A8}$$

The social security tax rate λ_{SS} is also determined in the model by the pension replacement ratio $\mathbb{S}$, the permanent income ratio, the relative population size of the retired and those of working age, and the aggregate employment rate.

$$\lambda_{SS} \sum_{\tau=1}^{T} \mu_\tau G_\tau (1 - \Pi^{\mho}) = \mathbb{S} \sum_{\tau=T+1}^{L} \mu_\tau G_\tau \tag{A9}$$

Stationary equilibrium. In the absence of the aggregate risk, I focus on the stationary equilibrium of the economy (StE). The stationary equilibrium of the model consists of consumption and savings policies $c(x, \tau)$, $a(x, \tau)$ as well as constant production factor prices, including the real interest rate R and the wage W, the initial wealth of newborns b_1, the unemployment benefit ζ, the tax rate λ, and the time-invariant distribution $(\psi_1, \psi_2, \ldots, \psi_L)$ such that

1. Consumption and savings policies are optimal, given the real interest rate R, the wage W, and the tax rate λ:

$$\begin{aligned} c(x, \tau) &= c^*(x, \tau) \\ a(x, \tau) &= a^*(x, \tau) \end{aligned} \tag{A10}$$

2. Distributions $(\psi_1, \psi_2, \ldots, \psi_L)$ are consistent with optimizing household behaviors, as described in Equation 24.

3. The factor markets are clearing.

$$\begin{aligned} &\sum_{\tau} \mu_\tau \int_X a(x, \tau) \mathrm{d}\psi_\tau = K \\ &\sum_{\tau=1}^{T} \mu_\tau \Pi_\tau^E = N \end{aligned} \tag{A11}$$

4. Firm optimization under competitive factor markets.

$$\begin{aligned} W &= Z(1 - \alpha)(K/N)^{\alpha} \\ R &= 1 + Z\alpha(K/N)^{\alpha - 1} - \delta_K \end{aligned} \tag{A12}$$

5. The initial bank balances of newborns are equal to their accidental bequests

$$b_1 = \sum_{\tau} \mu_\tau D \int_{x \in X} a(x, \tau) R \mathrm{d}\psi_\tau \tag{A13}$$

and

6. The government budget is balanced as described in Equations A8 and A9.

The economy attains different stationary equilibria, depending on the specific assumptions about the size and heterogeneity of income risks Γ. In the subjective model,

the stationary equilibrium further depends jointly on $\tilde{\Gamma}$ and Γ and their population distribution.

Results. Figure A.11 presents a comparison between the baseline model and the subjective model in the stationary equilibrium, alongside empirical moments computed under a redefined measure of wealth. Specifically, since savings serve as productive capital in general equilibrium, I use net worth from the SCF—rather than net liquid wealth—to benchmark the general-equilibrium outcomes. Under this definition, the share of H2M households based on net wealth falls to 14%, which is lower than when wealth is measured only by liquid assets. The wealth Gini coefficient falls to 0.82, indicating a slightly reduced degree of underlying inequality relative to the liquid wealth Gini. The average wealth-to-income ratio increases to 5.45, a substantially higher value than the corresponding measure based solely on liquid wealth.

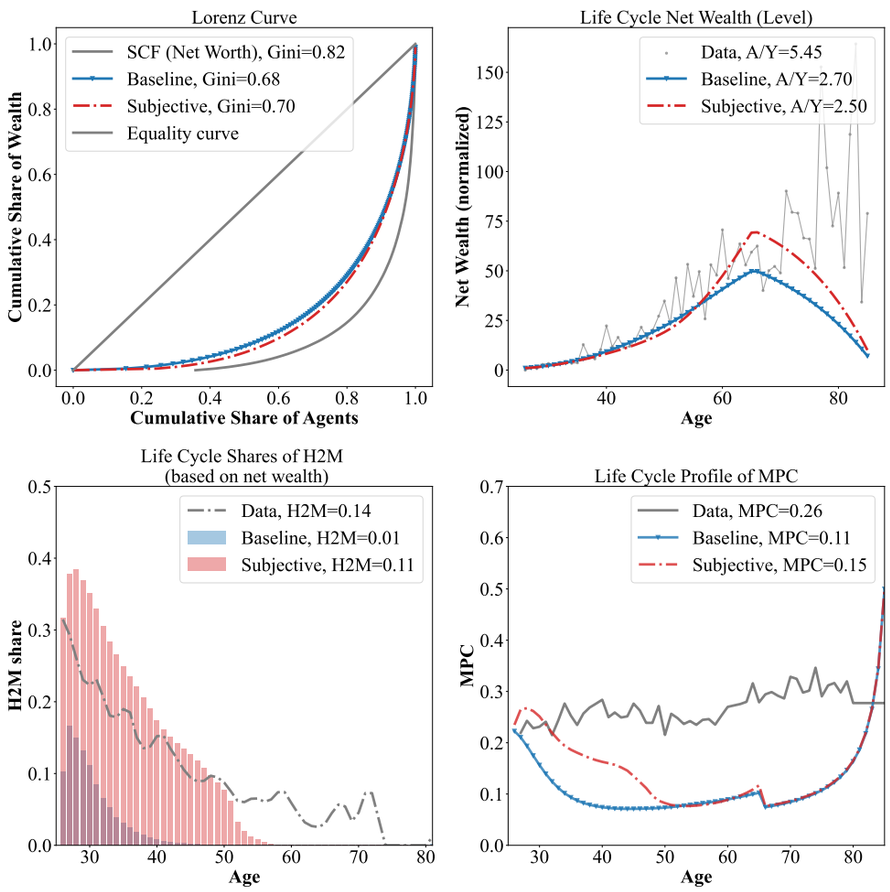
Revisiting the comparison between the baseline and subjective models, the general-equilibrium feedbacks appear to attenuate, to varying extents, the effects present in the subjective model under partial equilibrium. Nonetheless, even with this attenuation, the subjective model stays much closer to the redefined data moments. For example, the average H2M share in the subjective model is 0.11—10 percentage points higher than in the baseline model—approaching its data counterpart of 0.14. The life-cycle profile of the H2M share is also largely consistent with the empirical pattern, though the model overstates the proportion of H2M households among the young. The wealth Gini is 2 percentage points closer to its empirical counterpart as well. At the same time, the average wealth-to-income ratio falls short of the baseline value. In both specifications, the resulting mean wealth is too low to match the data, a well-known limitation of the one-asset framework (see Kaplan and Violante (2022) for further discussion).

In general equilibrium, savings serve as productive capital, so both models must clear the same aggregate capital market at an endogenous interest rate. The mechanisms of the subjective model that dampen saving in partial equilibrium jointly lower its desired capital relative to the baseline; the equilibrium interest rate then adjusts to restore market clearing, inducing households to save more similarly and drawing the wealth distribution toward the baseline.

This attenuation, however, should not be read as evidence that general equilibrium broadly dampens the implications of subjective risk perceptions for inequality. It is specific to the one-asset framework adopted here, which imposes the strong assumption that all liquid savings are channeled into productive capital. In a setting that distinguishes liquid wealth from the aggregate capital stock, this price feedback on the wealth distribution would be considerably weaker. Crucially, the subjective model's improvements over the baseline in the H2M share and the average MPC persist in general equilibrium, so its central implications for hand-to-mouth behavior and consumption responsiveness remain intact even as the inequality gap narrows.

Table A.6 reports the key moments from the same list of alternative model assumptions in comparison with the data moments calculated based on net worth in the SCF. Even in general equilibrium, the improvement in the model's fit with subjective risk beliefs is robust to a list of considerations.

FIGURE A.11. Baseline versus subjective in the general-equilibrium model



Note: The figure displays model-implied moments alongside their empirical counterparts, for both the baseline and the subjective model in the general equilibrium. Shown are: the Lorenz curve of household wealth (upper left); life-cycle profiles of average wealth normalized by its value at age 25 (upper right); the share of H2M households over the life cycle (bottom left); and reported marginal propensities to consume (MPCs) out of a hypothetical $500 windfall (bottom right). All wealth statistics are based on total net worth from the 2016 SCF, and reported MPCs are obtained from the SCE.

TABLE A.4. The size and nature of idiosyncratic income risks in the literature

	σ_ψ	σ_θ	υ	E	Earning Process	Unemployment	Source
Huggett (1996)	[0.21,+]	N/A	N/A	N/A	AR(1)	No	Page 480
Krusell and Smith (1998)	N/A	N/A	[0.04,0.1]	[0.9,0.96]	N/A	Persistent	Page 876
Cagetti (2003)	[0.264, 0.348]	N/A	N/A	N/A	Random +MA innovations	No	Page 344
Gourinchas and Parker (2002)	[0.108,0.166]	[0.18, 0.256]	0.003	0.997	Permanent +transitory	Transitory	Table 1
Meghir and Pistaferri (2004)	0.173	[0.09, 0.21]	N/A	N/A	Permanent +MA	No	Table 3
Storesletten et al. (2004)	[0.094, +]	0.255	N/A	N/A	Persistent + transitory	No	Table 2
Blundell et al. (2008)	[0.1,+]	[0.169,+]	N/A	N/A	Permanent + MA	No	Table 6
Low et al. (2010)	[0.095,0.106]	0.08	0.028	N/A	Permanent+transitory with job mobility	Persistent	Table 1
Kaplan and Violante (2014)	0.11	N/A	N/A	N/A	Persistent	No	Page 1220
Krueger et al. (2016)	[0.196,+]	0.23	[0.046,0.095]	[0.894,0.95]	Persistent +transitory	Persistent	Page 26
Carroll et al. (2017)	0.10	0.10	0.07	0.93	Permanent+transitory	Transitory	Table 2
Bayer et al. (2019)	0.148	0.693	N/A	N/A	Persistent time+MA	No	Table 1
My Estimates based on SIPP	0.10	0.025	N/A	N/A	Permanent +transitory	No	Table 1

Note: the conservative (lower-bound) estimates/parameterizations on idiosyncratic income risks at annual frequencies, seen in the literature.

TABLE A.5. Model results: robustness

Model/Data	Gini	Bottom 0.9	Bottom 0.7	Bottom 0.5	Average A/Y	H2M share	Average MPC
Data	0.89	0.15	0.04	0.01	0.74	0.35	0.26
Baseline	0.73	0.40	0.14	0.05	1.11	0.09	0.17
Subjective	0.84	0.25	0.05	0.01	0.68	0.31	0.25
Baseline: $\sigma_\psi = \sigma_\theta = 0.1$	0.71	0.44	0.16	0.05	0.91	0.12	0.19
Subjective: $\sigma_\psi = \sigma_\theta = 0.1$	0.83	0.26	0.05	0.01	0.64	0.33	0.25
Baseline: $\zeta = 0.15$	0.65	0.47	0.21	0.09	1.48	0.01	0.15
Subjective: $\zeta = 0.15$	0.81	0.29	0.08	0.02	0.78	0.23	0.24
Baseline: $\zeta = 0.6$	0.74	0.39	0.13	0.04	1.06	0.12	0.18
Subjective: $\zeta = 0.6$	0.85	0.24	0.05	0.01	0.67	0.32	0.25
Subjective: $\kappa = 0.1$	0.81	0.31	0.07	0.02	0.63	0.30	0.25
Subjective: $\kappa = 10$	0.85	0.23	0.05	0.01	0.70	0.31	0.25
Subjective: $\mathbb{G}$, 5 equiprobable	0.83	0.26	0.06	0.01	0.70	0.30	0.24
Subjective: $\mathbb{G}$, Gauss-Hermite 3	0.82	0.28	0.07	0.01	0.74	0.29	0.24
Subjective: $\mathbb{G}$, Gauss-Hermite 5	0.83	0.27	0.06	0.01	0.71	0.30	0.24

Note: The table shows, under various alternative values of true wage risks, the unemployment insurance replacement ratio and permanent-to-transitory ratios, the model-implied Gini coefficients, the wealth shares owned by the bottom 90, 70, and 50 percent of agents, the mean wealth-to-income ratio, the shares of hand-to-mouth agents (H2M), and average MPC in the partial-equilibrium stationary distribution. H2M is defined as those whose liquid wealth is no more than two weeks' (1/24 of annual) income. MPC is calculated in the model as the proportion of a hypothetical one-time $500 windfall spent, as measured in the SCE. Other wealth statistics in the data are computed for net liquid wealth in the 2016 SCF.

TABLE A.6. Summary of model results and data moments in the general equilibrium

Model/Data	Gini	Bottom 0.9	Bottom 0.7	Bottom 0.5	Average A/Y	H2M share	Average MPC
SCF (net worth)	0.82	0.28	0.08	0.02	5.45	0.14	0.26
Baseline	0.68	0.46	0.19	0.07	2.70	0.01	0.11
Subj. Wage (Lower)	0.68	0.47	0.18	0.06	2.54	0.04	0.13
Subj. Wage (Uniform)	0.71	0.43	0.16	0.06	2.48	0.06	0.14
Subj. Wage	0.69	0.46	0.18	0.06	2.54	0.06	0.14
Subj. UE	0.69	0.45	0.18	0.07	2.62	0.04	0.12
Subjective	0.70	0.45	0.17	0.05	2.50	0.11	0.15
Subj. Wage + Obj. Het. UE	0.70	0.45	0.17	0.05	2.63	0.11	0.15
Obj. Het. Wage + Subj. UE	0.65	0.53	0.21	0.07	2.38	0.10	0.15
Obj. Het. Both	0.65	0.53	0.21	0.07	2.51	0.10	0.15
Biased Expectation	0.69	0.48	0.17	0.06	2.55	0.10	0.15
Fanning	0.70	0.46	0.17	0.05	2.52	0.10	0.15
Hump	0.70	0.45	0.16	0.05	2.57	0.12	0.16
Right Skew	0.71	0.45	0.16	0.05	2.55	0.12	0.16
CRRA	0.75	0.38	0.13	0.04	2.36	0.14	0.16
Patience	0.71	0.45	0.16	0.05	2.47	0.11	0.15
Bequests	0.71	0.44	0.15	0.05	2.42	0.11	0.15
Borrowing	0.72	0.44	0.15	0.04	2.50	0.18	0.14

Note: The table shows the model-implied Gini coefficients, the wealth shares owned by the bottom 90, 70, and 50 percent of agents, the mean wealth-to-income ratio, the shares of hand-to-mouth agents (H2M), and average MPC in the general-equilibrium stationary distribution under various model specifications in the general equilibrium. H2M is defined as those whose net worth is no more than two weeks' (1/24 of annual) income. MPC is calculated in the model as the proportion of a hypothetical one-time $500 windfall spent, as measured in the SCE. Other wealth statistics in the data are computed for net worth in the 2016 SCF.